



CSC503 x SENG747: Data Mining
Fall 2026

Photo: Steve McCurry

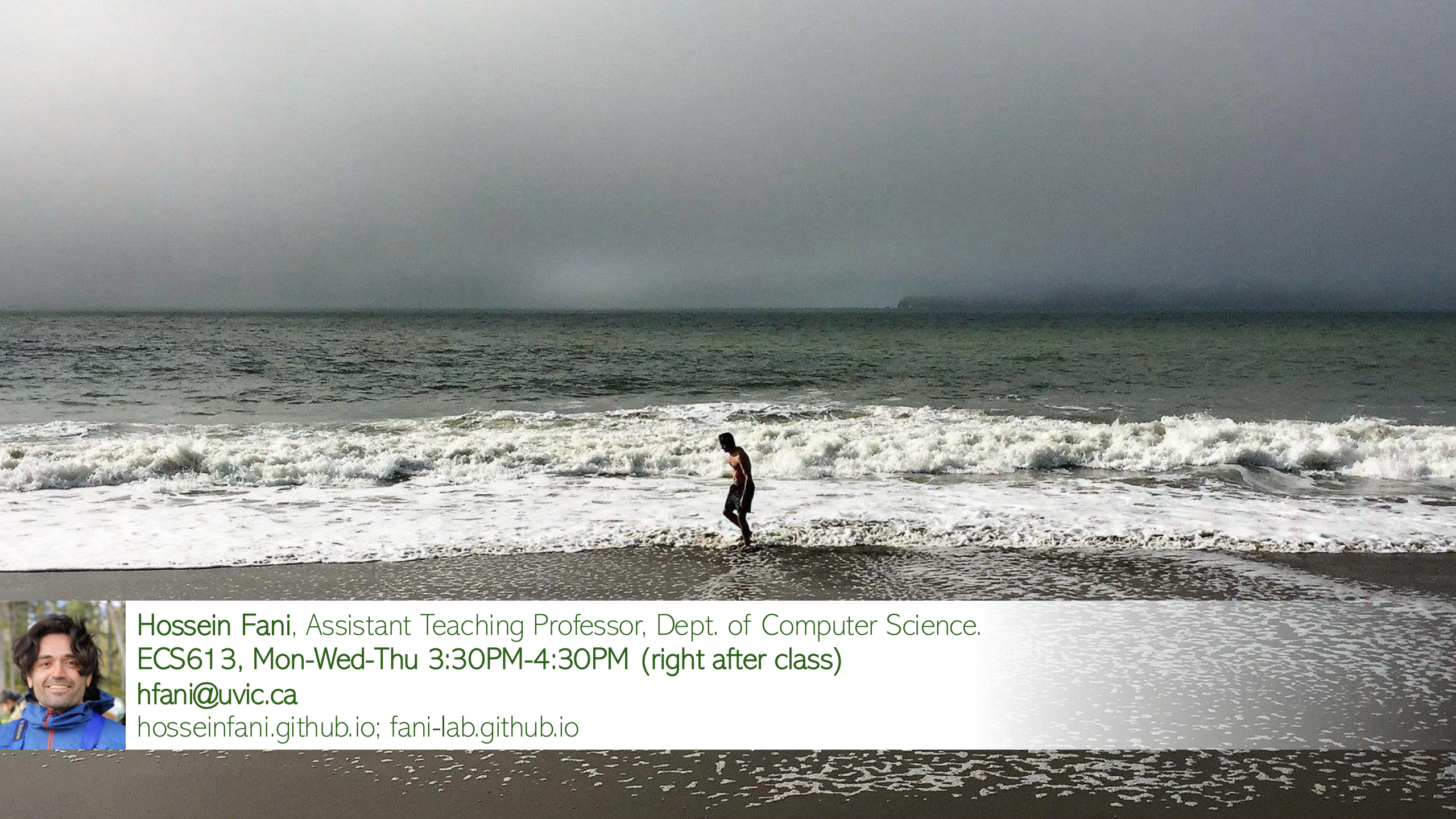


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Undergrad (80.8%) and Graduate (19.2%) Students

Undergraduate (97): Computer Science (61), Software Engineering (12), Psychology (6), Math (4), Data Science (3), Geomatics (2), Health Information Science (1), Non-Degree (3)

Graduate (23): Computer Science (8), Telecoms & Information Security (8), Applied Data Science (5), Electrical & Computer Engineering (1), Management (1)



Hossein Fani, Assistant Teaching Professor, Dept. of Computer Science.
ECS613, Mon-Wed-Thu 3:30PM-4:30PM (right after class)

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Outline: Week 01 A: Monday, B: Wednesday, C: Thursday

I) An Odyssey

II) Artificial Intelligence (AI)

III) State of AI

III) Data Mining

Outline: Week 02 A: Monday, Course Logistics & Topics

Outline: W01 B

I) An Odyssey

II) Artificial Intelligence (AI)

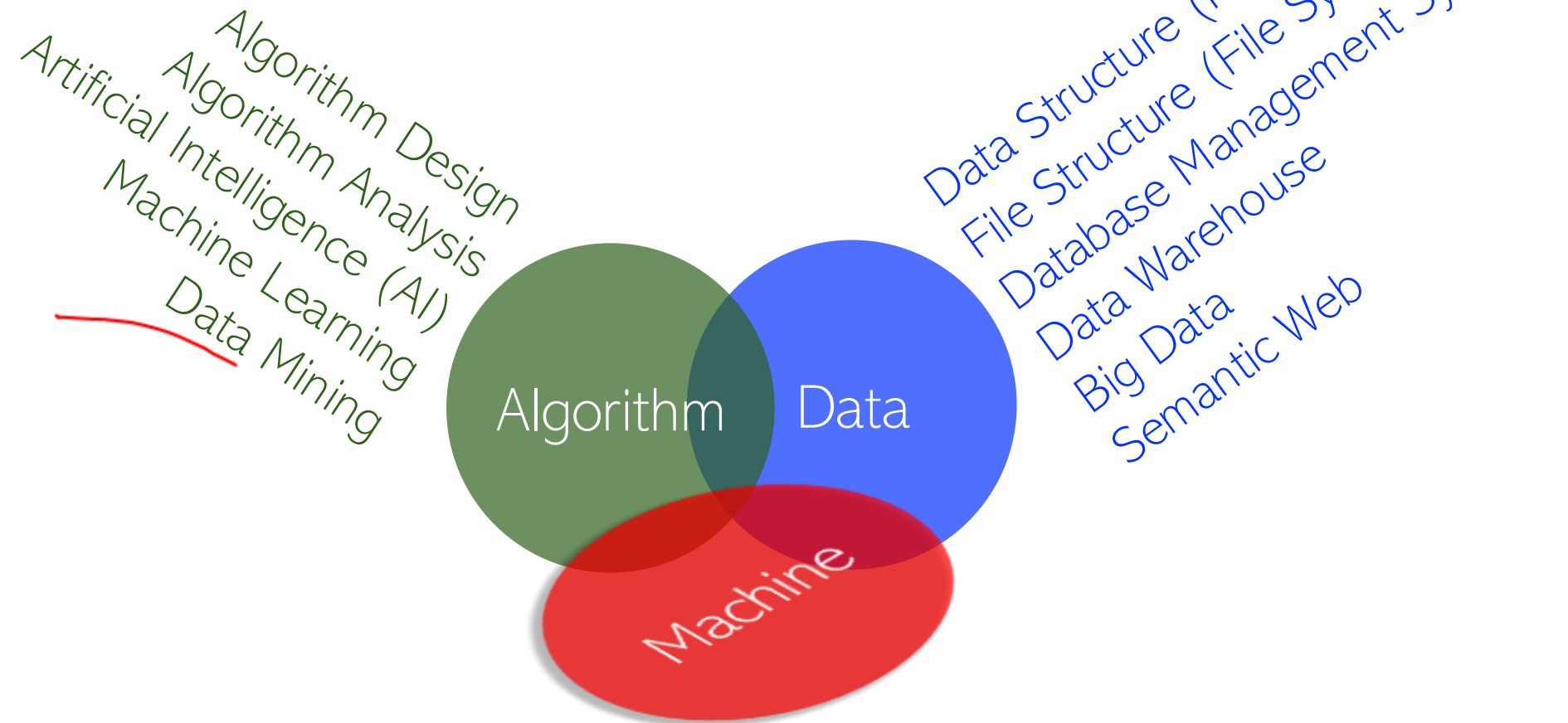
III) State of AI

III) Data Mining

Outline: Week 02 A: Monday, Course Logistics & Topics

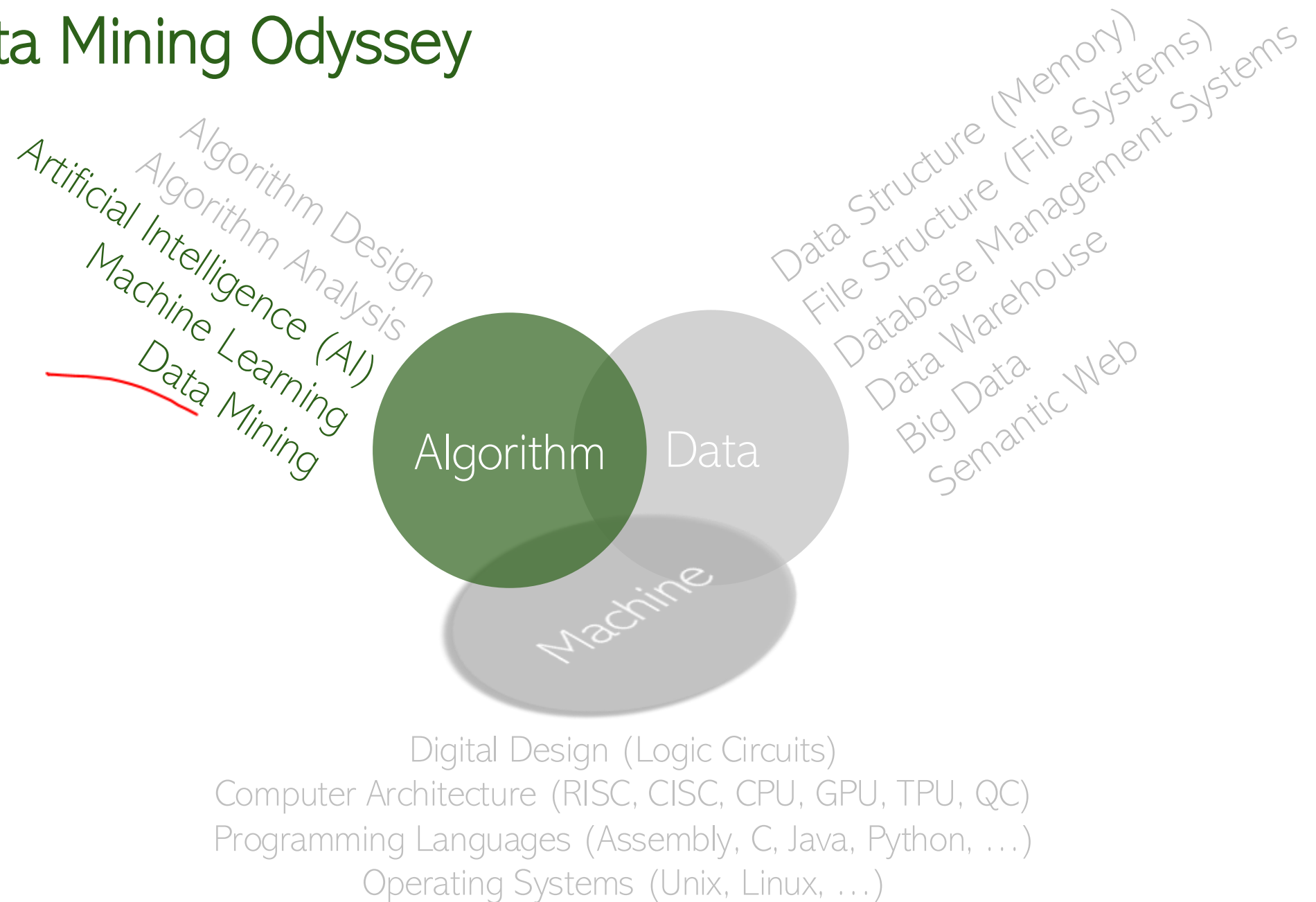
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A Data Mining Odyssey



Digital Design (Logic Circuits, qbits)
Computer Architecture (RISC, CISC, CPU, GPU, TPU, QC)
Programming Languages (Assembly, C, Java, Python, ...)
Operating Systems (Unix, Linux, ...)

7 A Data Mining Odyssey



Outline: W01 B

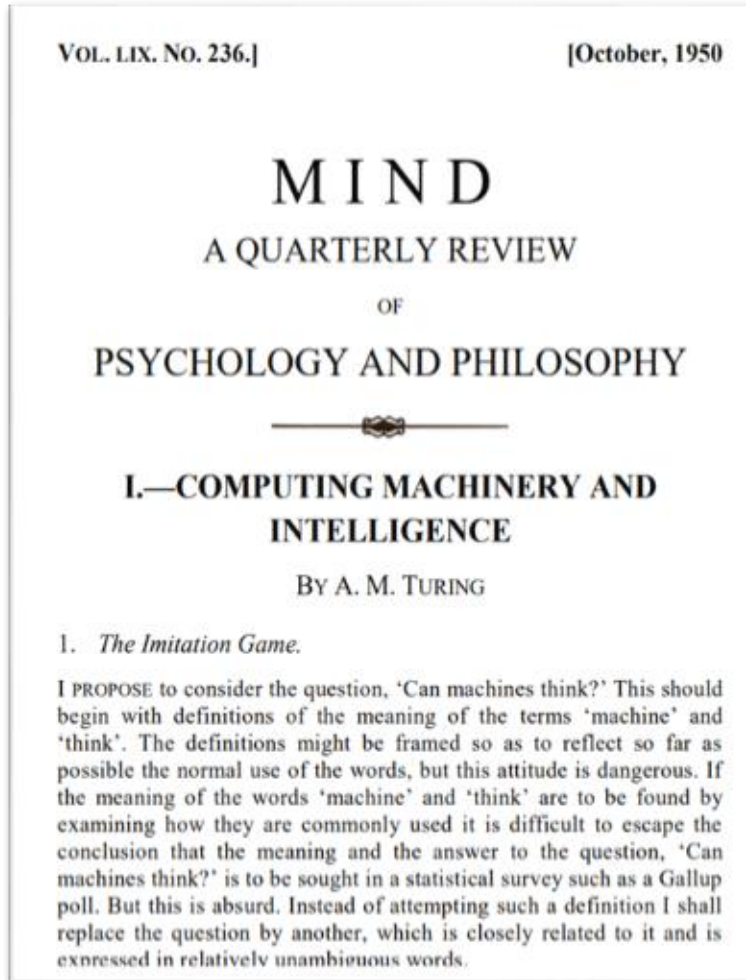
I) An Odyssey

II) Artificial Intelligence (AI)

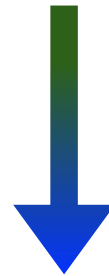
III) State of AI

III) Data Mining

Forms of Intelligence



Human-level



From Processing to Understanding to Communication
Text, Speech, Visual, Cognition, Reason

Intelligence = Emulate Human: Imitation Game

Alan Mathison Turing (/ˈtʃʊərɪŋ/; 1912 – 1954)

English mathematician, computer scientist, logician, cryptanalyst, philosopher and theoretical biologist.

Widely considered to be the father of theoretical computer science.

Influential in theoretical computer science, formalisation of the concepts of algorithm and computation with the Turing machine.

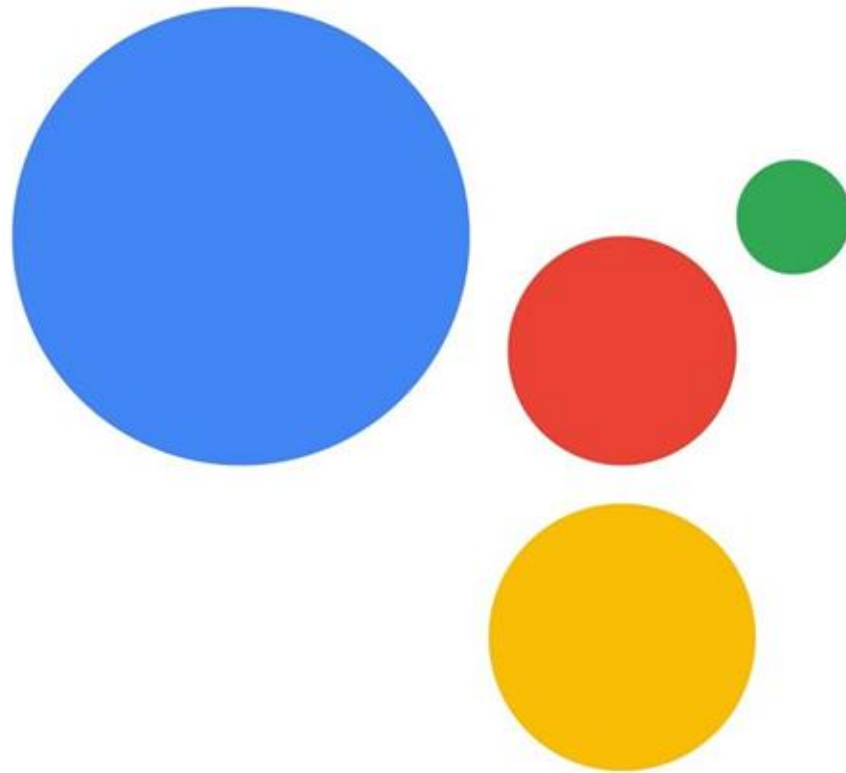


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Artificial Intelligence: Turing Test

2018: A Conversation with Google Duplex!

Sundar Pichai, Google's CEO: <https://www.youtube.com/watch?v=D5VN56jQMWM>



Artificial Intelligence: Turing Test is Misleading

*"The ability to **fake intelligence** does not amount to having intelligence."*

John Searle, Philosopher, "Chinese Room" argument to challenge Turing's imitation game, 1980.

*"Human-Level Intelligence or **Animal-Like Abilities**?"*

Adnan Darwiche, University of California.

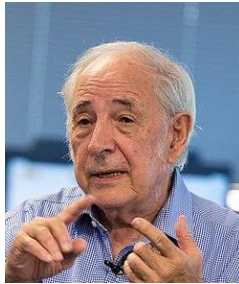
On the Dangers of **Stochastic Parrots**: Can Language Models Be Too Big? 🦜

Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell, FAccT 2021

The Book of **Why**: The New Science of **Cause and Effect**

The Ladder of Causation

Judea Pearl, 2018, Turing Award Winner



Artificial Intelligence: <https://openreview.net/forum?id=uyTL5Bvosj>

Published in Transactions on Machine Learning Research (05/2023)

Beyond the Imitation Game: Quantifying and extrapolating the capabilities of language models

Alphabetic author list:*

Aarohi Srivastava, Abhinav Rastogi, Abhishek Rao, Abu Awal Md Shoeb, Abubakar Abid, Adam Fisch, Adam R. Brown, Adam Santoro, Aditya Gupta, Adrià Garriga-Alonso, Agnieszka Kluska, Aitor Lewkowycz, Akshat Agarwal, Alethea Power, Alex Ray, Alex Warstadt, Alexander W. Kocurek, Ali Safaya, Ali Tazarv, Alice Xiang, Alicia Parrish, Allen Nie, Aman Hussain, Amanda Askell, Amanda Dsouza, Ambrose Slone, Ameet Rahane, Anantharaman S. Iyer, Anders Andreassen, Andrea Madotto, Andrea Santilli, Andreas Stuhlmüller, Andrew Dai, Andrew La, Andrew Lampinen, Andy Zou, Angela Jiang, Angelica Chen, Anh Vuong, Animesh Gupta, Anna Gottardi, Antonio Norelli, Anu Venkatesh, Arash Ghahramani, Arif Tabassum, Arul Menezes, Arun Kirubakaran, Asher Mullokkandov, Ashish Sabharwal, Austin Herrick, Avia Efrat, Aykut Erdem, Ayta Karakas, B. 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Reviewed on OpenReview: <https://openreview.net/forum?id=uyTL5Bvosj>

442 vs. 1



VOL. LIX, No. 236.]

[October, 1959]

MIND A QUARTERLY REVIEW OF PSYCHOLOGY AND PHILOSOPHY

I.—COMPUTING MACHINERY AND INTELLIGENCE

By A. M. TURING

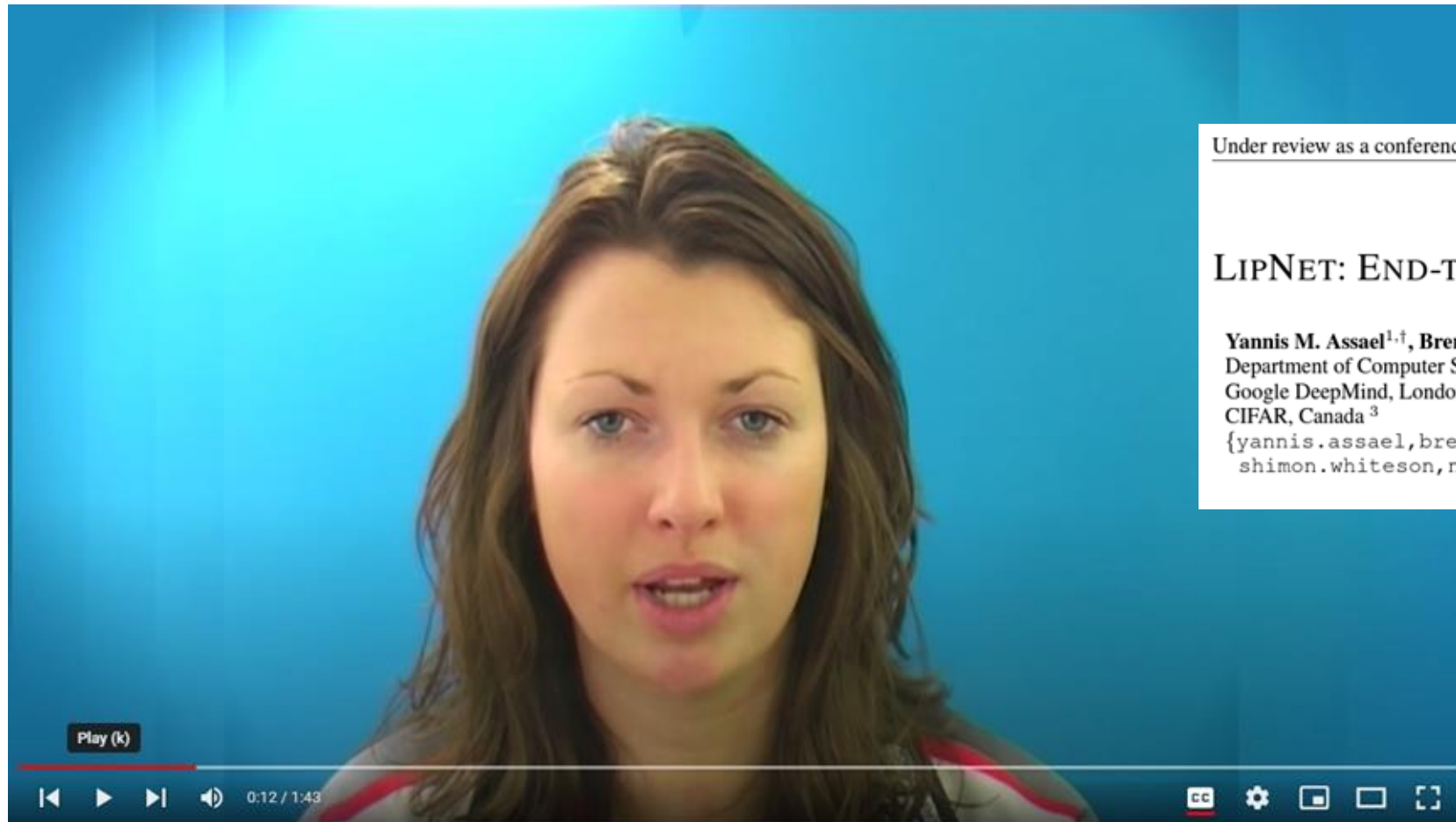
1. *The Imitation Game.*

I PROPOSE to consider the question, 'Can machines think?' This should begin with definitions of the meaning of the terms 'machine' and 'think'. The definitions might be framed so as to reflect so far as possible the normal use of the words, but this attitude is dangerous. If the meaning of the words 'machine' and 'think' are to be found by examining how they are commonly used it is difficult to escape the conclusion that the meaning and the answer to the question, 'Can machines think?' is to be sought in a statistical survey such as a Gallup poll. But this is absurd. Instead of attempting such a definition I shall replace the question by another, which is closely related to it and is expressed in relatively unambiguous words.

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Artificial Intelligence: Exceed Human Processing

<https://www.youtube.com/watch?v=fa5QGremQf8&list=PLXkuFIFnXUAPlrXKgtlpctv2NuSo7xw3k&index=1>



Under review as a conference paper at ICLR 2017

LIPNET: END-TO-END SENTENCE-LEVEL LIPREADING

Yannis M. Assael^{1,†}, Brendan Shillingford^{1,†}, Shimon Whiteson¹ & Nando de Freitas^{1,2,3}
Department of Computer Science, University of Oxford, Oxford, UK ¹
Google DeepMind, London, UK ²
CIFAR, Canada ³
{yannis.assael,brendan.shillingford,
shimon.whiteson,nando.de.freitas}@cs.ox.ac.uk

LipNet: How easy do you think lipreading is?

Artificial Intelligence: Exceed Human Cognition

AlphaTensor: An Algorithm Designs an Algorithm
No Need for Human Brain Work!

Discovering faster matrix multiplication algorithms with reinforcement learning

[Alhussein Fawzi](#) , [Matej Balog](#), [Aja Huang](#), [Thomas Hubert](#), [Bernardino Romera-Paredes](#), [Mohammadamin Barekatin](#), [Alexander Novikov](#), [Francisco J. R. Ruiz](#), [Julian Schrittwieser](#), [Grzegorz Swirszcz](#), [David Silver](#), [Demis Hassabis](#) & [Pushmeet Kohli](#)

[Nature](#) **610**, 47–53 (2022) | [Cite this article](#)

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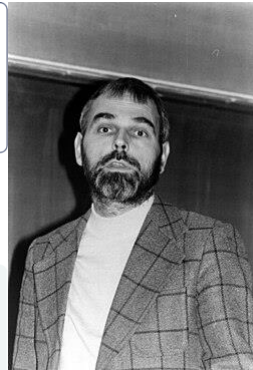
718k Accesses | **560** Citations | **3460** Altmetric | [Metrics](#)

matrix sizes. Particularly relevant is the case of 4×4 matrices in a finite field, where AlphaTensor's algorithm improves on **Strassen's two-level algorithm** for the first time, to our knowledge, since its discovery **50 years ago**². We further showcase the flexibility of AlphaTensor through different

Artificial Intelligence: Exceed Human Cognition

<https://deepmind.google/blog/discovering-novel-algorithms-with-alphatensor/>

$$\begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{bmatrix} \times \begin{bmatrix} b_{1,1} & b_{1,2} \\ b_{2,1} & b_{2,2} \end{bmatrix} = \begin{bmatrix} c_{1,1} & c_{1,2} \\ c_{2,1} & c_{2,2} \end{bmatrix}$$



Volker Strassen
(1936 - Present)
German mathematician
Professor emeritus
University of Konstanz

7 Multiplications
 $n \times n$ matrices: $(n^{\log_2 7}) \approx (n^{2.81})$

$$\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} & a_{1,4} & a_{1,5} \\ a_{2,1} & a_{2,2} & a_{2,3} & a_{2,4} & a_{2,5} \\ a_{3,1} & a_{3,2} & a_{3,3} & a_{3,4} & a_{3,5} \\ a_{4,1} & a_{4,2} & a_{4,3} & a_{4,4} & a_{4,5} \end{bmatrix} \times \begin{bmatrix} b_{1,1} & b_{1,2} & b_{1,3} & b_{1,4} & b_{1,5} \\ b_{2,1} & b_{2,2} & b_{2,3} & b_{2,4} & b_{2,5} \\ b_{3,1} & b_{3,2} & b_{3,3} & b_{3,4} & b_{3,5} \\ b_{4,1} & b_{4,2} & b_{4,3} & b_{4,4} & b_{4,5} \\ b_{5,1} & b_{5,2} & b_{5,3} & b_{5,4} & b_{5,5} \end{bmatrix} = \begin{bmatrix} c_{1,1} & c_{1,2} & c_{1,3} & c_{1,4} & c_{1,5} \\ c_{2,1} & c_{2,2} & c_{2,3} & c_{2,4} & c_{2,5} \\ c_{3,1} & c_{3,2} & c_{3,3} & c_{3,4} & c_{3,5} \\ c_{4,1} & c_{4,2} & c_{4,3} & c_{4,4} & c_{4,5} \end{bmatrix}$$

Standard algorithm

$$\begin{aligned} h_1 &= a_{1,1} b_{1,1} \\ h_2 &= a_{1,1} b_{1,2} \\ h_3 &= a_{1,2} b_{2,1} \\ h_4 &= a_{1,2} b_{2,2} \\ h_5 &= a_{2,1} b_{1,1} \\ h_6 &= a_{2,1} b_{1,2} \\ h_7 &= a_{2,2} b_{2,1} \\ h_8 &= a_{2,2} b_{2,2} \end{aligned}$$

$$\begin{aligned} c_{1,1} &= h_1 + h_3 \\ c_{1,2} &= h_2 + h_4 \\ c_{2,1} &= h_5 + h_7 \\ c_{2,2} &= h_6 + h_8 \end{aligned}$$

8 Multiplications
 $n \times n$ matrices: (n^3)

Strassen's algorithm

$$\begin{aligned} h_1 &= (a_{1,1} + a_{2,2})(b_{1,1} + b_{2,2}) \\ h_2 &= (a_{2,1} + a_{2,2})b_{1,1} \\ h_3 &= a_{1,1}(b_{1,2} - b_{2,2}) \\ h_4 &= a_{2,2}(-b_{1,1} + b_{2,1}) \\ h_5 &= (a_{1,1} + a_{1,2})b_{2,2} \\ h_6 &= (-a_{1,1} + a_{2,1})(b_{1,1} + b_{1,2}) \\ h_7 &= (a_{1,2} - a_{2,2})(b_{2,1} + b_{2,2}) \end{aligned}$$

$$\begin{aligned} c_{1,1} &= h_1 + h_4 - h_5 + h_7 \\ c_{1,2} &= h_3 + h_5 \\ c_{2,1} &= h_2 + h_4 \\ c_{2,2} &= h_1 - h_2 + h_3 + h_6 \end{aligned}$$

$$\begin{aligned} h_1 &= a_{1,1}(-b_{2,1} - b_{2,2} - b_{2,3}) \\ h_2 &= (a_{1,2} + a_{2,2} - a_{2,3})(-b_{2,1} - b_{2,2}) \\ h_3 &= (a_{1,2} + a_{2,2} + a_{2,3})(-b_{2,1} - b_{2,2}) \\ h_4 &= (a_{1,2} + a_{2,2} + a_{2,3})(-b_{2,1} - b_{2,2}) \\ h_5 &= (-a_{1,2} - a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_6 &= (-a_{1,2} - a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_7 &= (a_{1,2} - a_{2,2})(b_{2,1} + b_{2,2}) \\ h_8 &= (a_{1,2} - a_{2,2})(b_{2,1} + b_{2,2}) \\ h_9 &= (-a_{1,2} - a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{10} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{11} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{12} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{13} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{14} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{15} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{16} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{17} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{18} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{19} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{20} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{21} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{22} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{23} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{24} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{25} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{26} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{27} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{28} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{29} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{30} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{31} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{32} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{33} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{34} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{35} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{36} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{37} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{38} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{39} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{40} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{41} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{42} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{43} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{44} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{45} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{46} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{47} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{48} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{49} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{50} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{51} &= (a_{1,2} + a_{2,2} - a_{2,3})(b_{2,1} + b_{2,2}) \\ h_{52} &= (a_{1,2} + a_{2,2} - 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"if the traditional algorithm taught in school multiplies a 4x5 by 5x5 matrix using 100 multiplications, and this number was reduced to 80 with human ingenuity, AlphaTensor has found algorithms that do the same operation using just 76 multiplications."

Strassen's 2x2 algorithm recursively twice for a 4x4 matrix: 7 x 7 = 49 Multiplications
AlphaTensor: 47 Multiplications

Artificial Intelligence: **Back to Human** but not Just Observation

Move Beyond Processing, Understanding, Emotion, Cognition, Communication



Stephen Gary Wozniak (/ˈwɒzniæk/; 1950 - Present), “Woz”, American electronics engineer, computer programmer, philanthropist, inventor, & technology entrepreneur. In 1976, with Steve Jobs, co-founded Apple Computer.
<http://woz.org/>



Ben Goertzel
Cognitive scientist, AI researcher, CEO and founder of SingularityNET, leader of OpenCog Foundation, and AGI Society, and chair of Humanity+. Helps popularize Artificial General Intelligence.
<https://goertzel.org/>



Nils John Nilsson (1933 – 2019)
American computer scientist and founding researcher in AI.
Known for his contributions to search, planning, knowledge representation, and robotics.
<http://robotics.stanford.edu/~nilsson/>

Artificial **General** Intelligence: **All-in-One** Anything Human Can **Do**

From Observation Toward **Intervention** (Action, Robotics)



Wozniak: Coffee Test

A **machine enters** a home and make a coffee: find coffee machine, find coffee, add water, find a mug, brew the coffee (pushing proper buttons).

Order matters: let me tell you a related funny, but costly, personal story on how I killed my iphone6 for my computer vision course project.



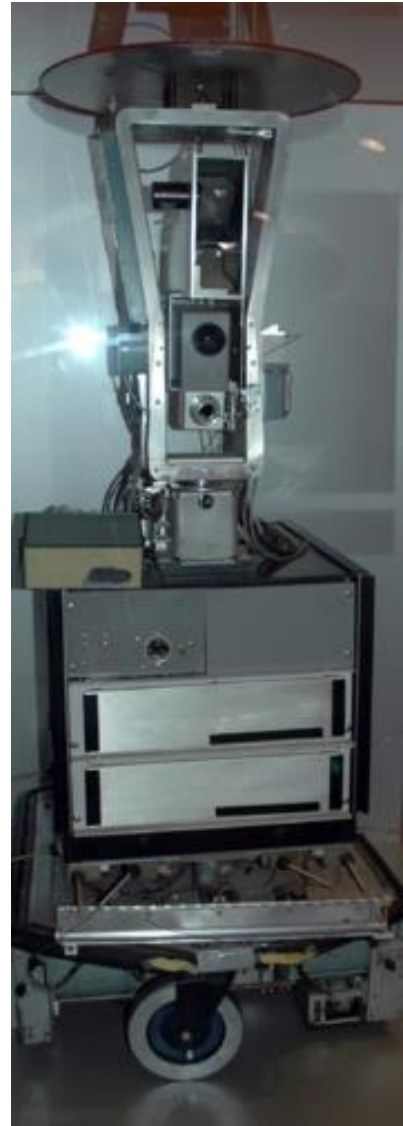
Goertzel: Robot College Student Test

A **machine enrolls** in a university, taking and passing the same classes that humans would, and obtaining a degree. *Not just theory, but also action: Drummer, Painter, Dancer, ...*



Nilsson: The Employment Test

A machine does an **economically important** job at least as well as humans in same job. Shakey (1966) the first general-purpose **robot** able to **reason about its own actions**.



Artificial Intelligence



RoboCop, 1987, Paul Verhoeven

Prior to Turing: Machine (hardware) was exceeding human physically in both running algorithms and interventions (robots).

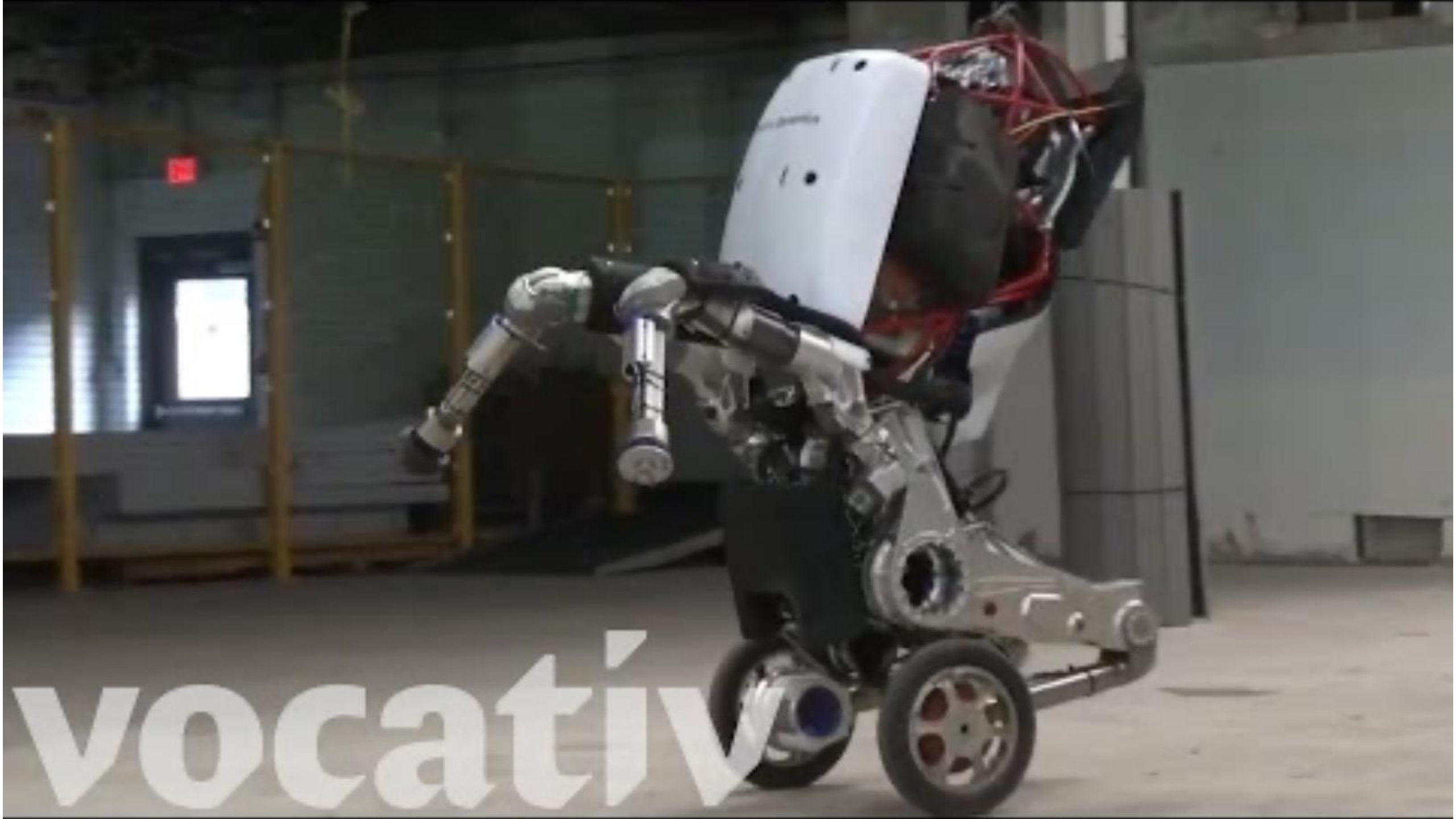


Robots: Exceeding Human **but only Physically**

Handle at Boston Dynamics. It can jump four feet in the air and zip around at nine miles per hour.

<https://www.youtube.com/watch?v=7h8mX9ZMs7g>





vocativ

Artificial Intelligence

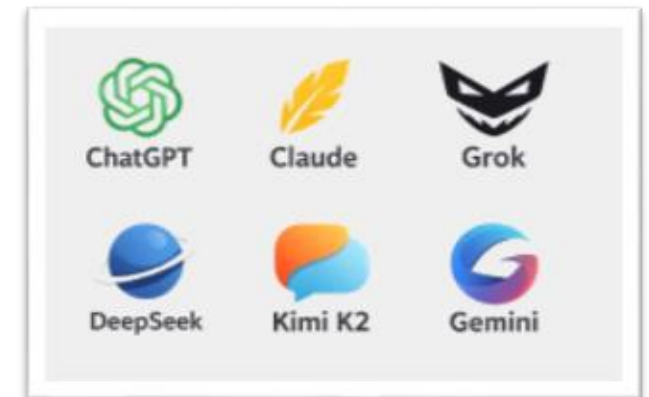


RoboCop, 1987, Paul Verhoeven

Prior to Turing: Machine (hardware) was exceeding human physically in both running algorithms and interventions (robots).



Prior to Wozniak: Algorithms (software) was exceeding human in observational study.
(Whether or not runs on a machine for efficiency. Any algorithm can be traced manually, be it a language model.)



Artificial Intelligence



RoboCop, 1987, Paul Verhoeven

Prior to Turing: Machine (hardware) was exceeding human physically in both running algorithms and interventions (robots).

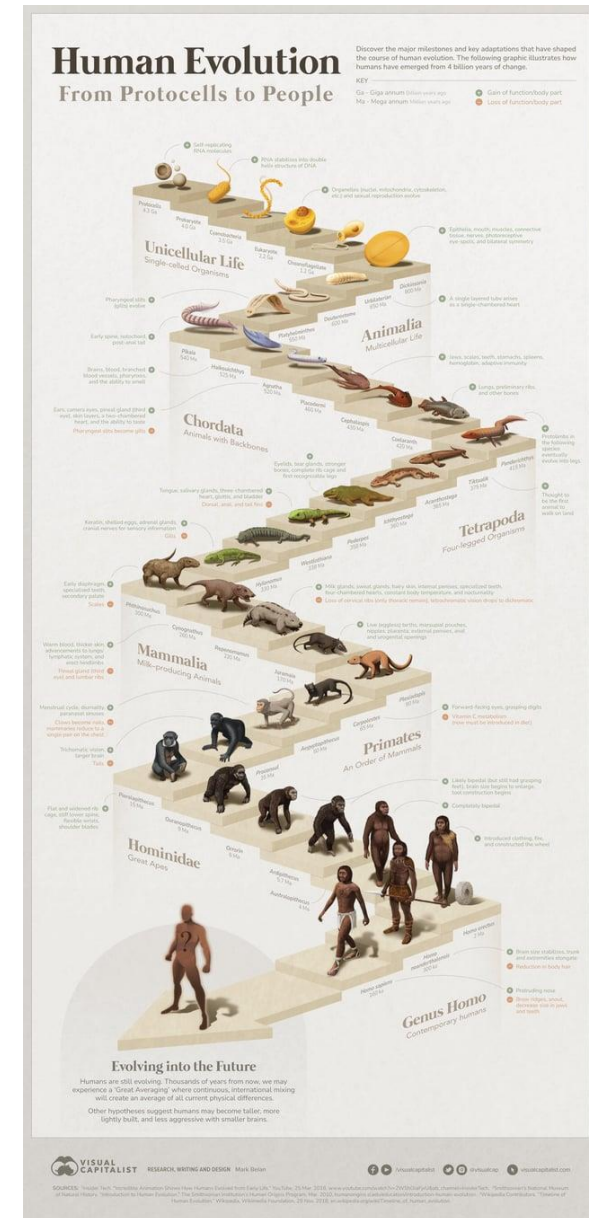


Prior to Wozniak: Algorithms (software) was exceeding human in observational study.



Post- Wozniak: AI plugs into robots **to intervene** in the world.
Non-biological Human = Humanoid

Ex Machina, 2015, Alex Garland



Artificial Intelligence: Physical Intelligence

2024: Google DeepMind Robotics Team

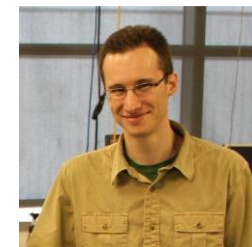
<https://twitter.com/i/status/1743328189802598861>

Sergey Levine

Professor at UC Berkeley

Robotics & machine learning

Co-founder of Physical Intelligence





Knife Scene, Ex Machina, 2015, Alex Garland: <https://www.youtube.com/watch?v=67VATPxULPk>

Deus Ex Machina /DAY uus + eks + MAA ki naa/ means "god from the machine." In ancient Greek theater, when actors playing gods carried onto stage by a machine. These gods would then serve as the ultimate arbiters of right and wrong and decide how the story ends. But this film is just called "*Ex Machina*" without the "*Deus*." A machine without a god. <https://www.looper.com/148401/the-ending-of-ex-machina-finally-explained/>



Artificial Intelligence!

Anthropic researcher warns AI *'could kill us all by the end of the decade'*

<https://www.youtube.com/watch?v=-dusfhk72HY>



Jacob Coxon, AI researcher
worked at Anthropic and OpenAI



Dario Amodei (1983-present)
American AI researcher and entrepreneur
In 2021, he and his sister Daniela Amodei co-founded Anthropic (Claude)
He was the vice president of research at OpenAI



Samuel Harris Altman (1985)
American entrepreneur & investor
CEO OpenAI.

Outline: W01C

I) An Odyssey

II) Artificial Intelligence (AI)

III) State of AI

III) Data Mining

State of AI Report

October 1, 2020

About the authors



Nathan Benaich

Nathan is the General Partner of **Air Street Capital**, a venture capital firm investing in AI-first technology and life science companies. He founded RAAIS and London.AI, which connect AI practitioners from large companies, startups and academia, and the RAAIS Foundation that funds open-source AI projects. He studied biology at Williams College and earned a PhD from Cambridge in cancer research.



Ian Hogarth

Ian is an **angel investor** in 60+ startups. He is a Visiting Professor at UCL working with Professor Mariana Mazzucato. Ian was co-founder and CEO of Songkick, the concert service used by 17M music fans each month. He studied engineering at Cambridge where his Masters project was a computer vision system to classify breast cancer biopsy images. He is the Chair of Phasecraft, a quantum software company.

Definitions

Artificial intelligence (AI): A broad discipline with the goal of creating intelligent machines, as opposed to the natural intelligence that is demonstrated by humans and animals. It has become a somewhat catch all term that nonetheless captures the long term ambition of the field to build machines that emulate and then exceed the full range of human cognition.

Machine learning (ML): A subset of AI that often uses statistical techniques to give machines the ability to "learn" from data without being explicitly given the instructions for how to do so. This process is known as "training" a "model" using a learning "algorithm" that progressively improves model performance on a specific task.

Reinforcement learning (RL): An area of ML concerned with developing software agents that learn goal-oriented behavior by trial and error in an environment that provides rewards or penalties in response to the agent's actions (called a "policy") towards achieving that goal.

Deep learning (DL): An area of ML that attempts to mimic the activity in layers of neurons in the brain to learn how to recognise complex patterns in data. The "deep" in deep learning refers to the large number of layers of neurons in contemporary ML models that help to learn rich representations of data to achieve better performance gains.

Definitions

Algorithm: An unambiguous specification of how to solve a particular problem.

Model: Once a ML algorithm has been trained on data, the output of the process is known as the model. This can then be used to make predictions.

Supervised learning: A model attempts to learn to transform one kind of data into another kind of data using labelled examples. This is the most common kind of ML algorithm today.

Unsupervised learning: A model attempts to learn a dataset's structure, often seeking to identify latent groupings in the data without any explicit labels. The output of unsupervised learning often makes for good inputs to a supervised learning algorithm at a later point.

Transfer learning: An approach to modelling that uses knowledge gained in one problem to bootstrap a different or related problem, thereby reducing the need for significant additional training data and compute.

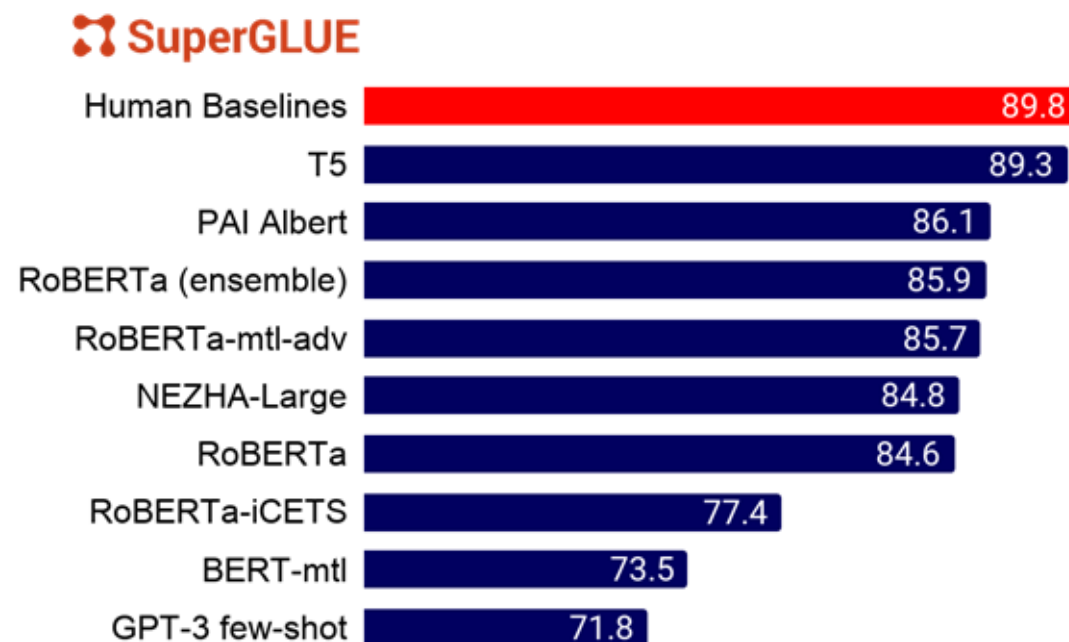
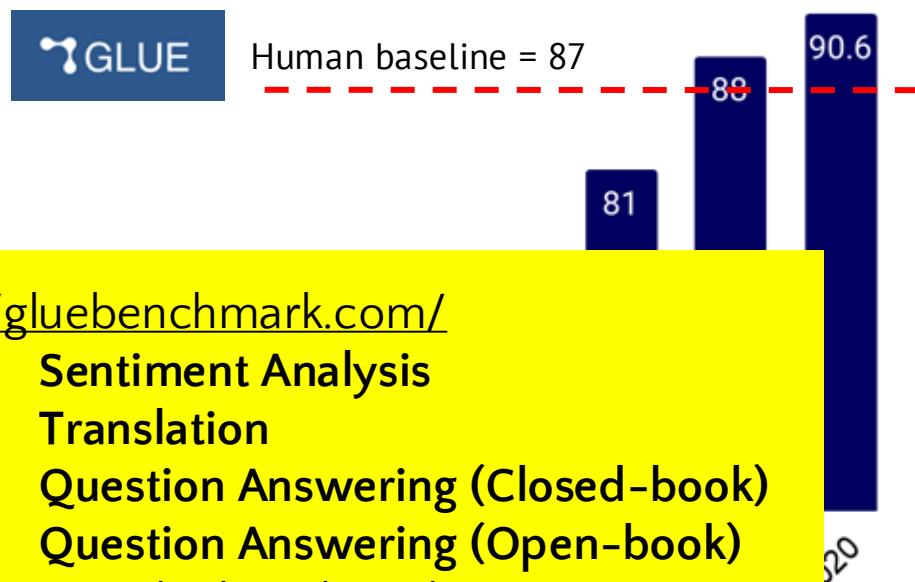
Natural language processing (NLP): Enabling machines to analyse, understand and manipulate language.

Computer vision: Enabling machines to analyse, understand and manipulate images and video.

NLP benchmarks take a beating: Over a dozen teams outrank the human GLUE baseline

► It was only 12 months ago that the human GLUE benchmark was beat by 1 point. Now SuperGLUE is in sight.

- GLUE (General Language Understanding Evaluation) and its more challenging sibling SuperGLUE are benchmarks that evaluate NLP systems at a range of tasks spanning logic, common sense understanding, and lexical semantics. The human benchmark on GLUE is reliably beat today and the SuperGLUE human benchmark is almost surpassed too!



<https://gluebenchmark.com/>

Sentiment Analysis

Translation

Question Answering (Closed-book)

Question Answering (Open-book)

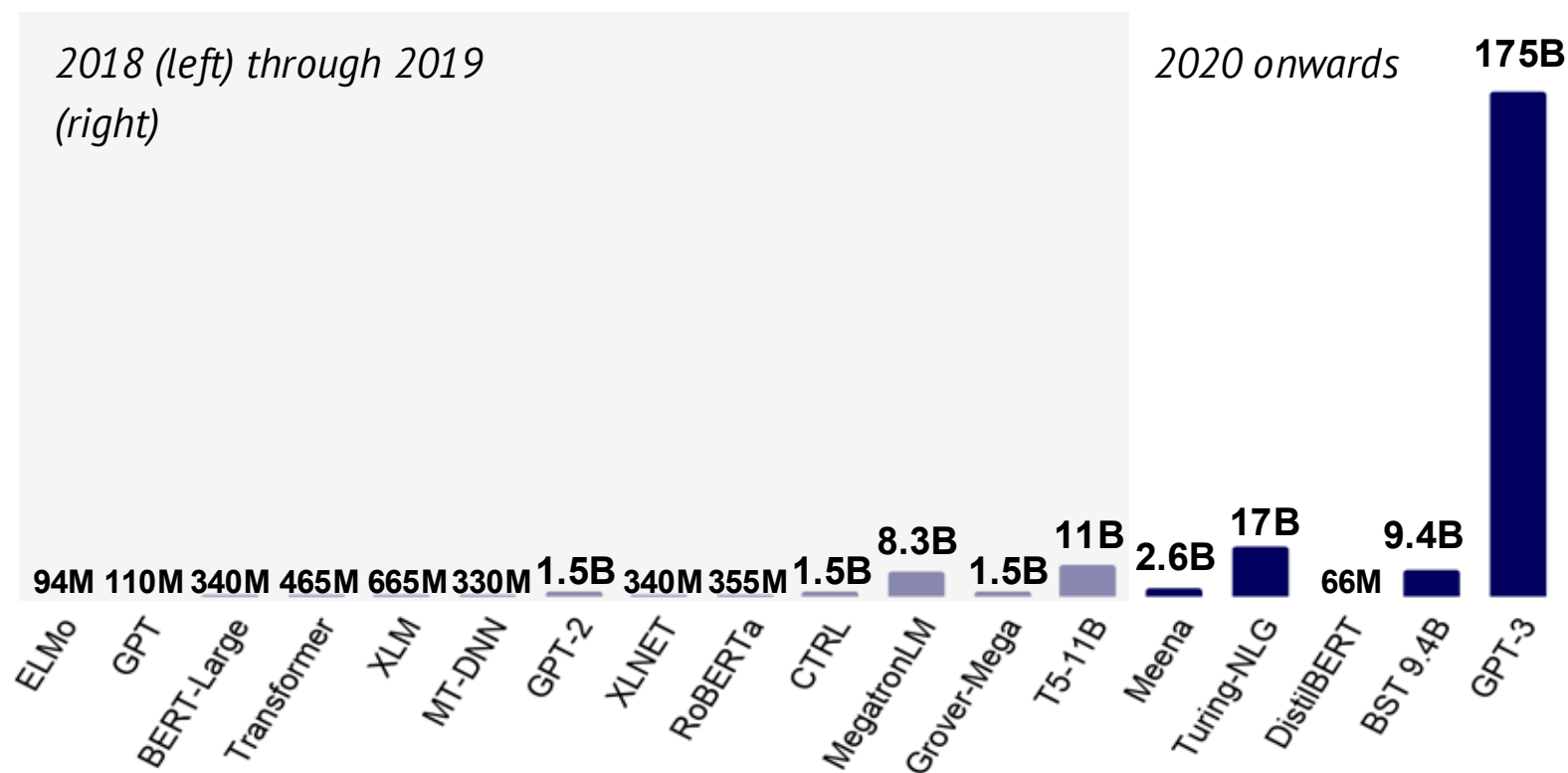
<https://super.gluebenchmark.com/>

Question Answering (Open-ended)

Cause/Effect

Language models: Welcome to the Billion Parameter club

- ▶ Huge models, large companies and massive training costs dominate the hottest area of AI today, NLP.



Note: The number of parameters indicates how many different coefficients the algorithm optimizes during the training process.

State of AI Report

October 12, 2021

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One year after General Language Understanding Evaluation (GLUE), **SuperGLUE is solved**

► 3 different teams from Baidu, Google and Microsoft all surpass human baselines on the SuperGLUE NLP tasks.

- Baidu's ERNIE 3.0 is the best scoring model (90.6%), outperforming the human baseline by 0.8 percentage point.
- ERNIE 3.0 stands out from two perspectives: its pre-training data and its historical development.
- Data: In addition to a massive text corpus, ERNIE 3.0 uses a large-scale knowledge graph of 50 million

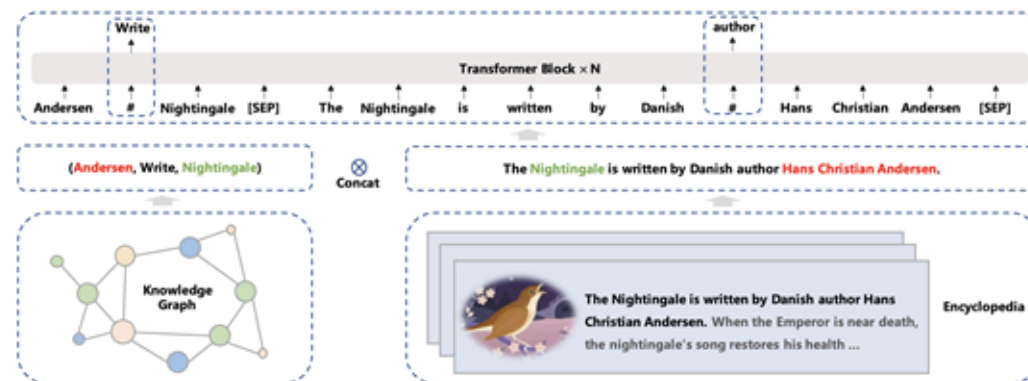
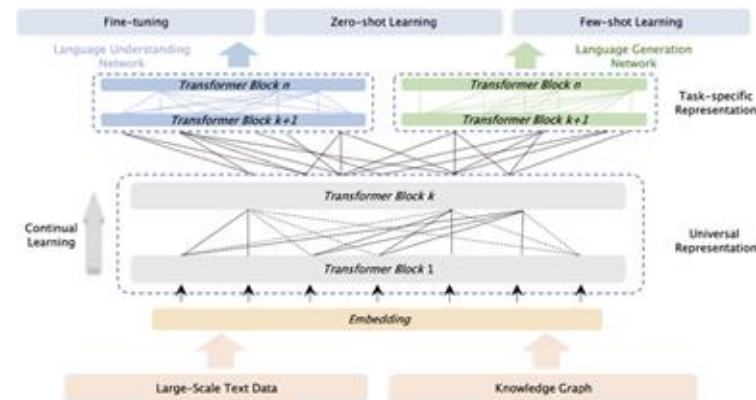


Figure: Pre-training using sentence re-ordering and knowledge graphs.



<https://gluebenchmark.com/>

Sentiment Analysis

Translation

Question Answering (Closed-book)

Question Answering (Open-book)

<https://super.gluebenchmark.com/>

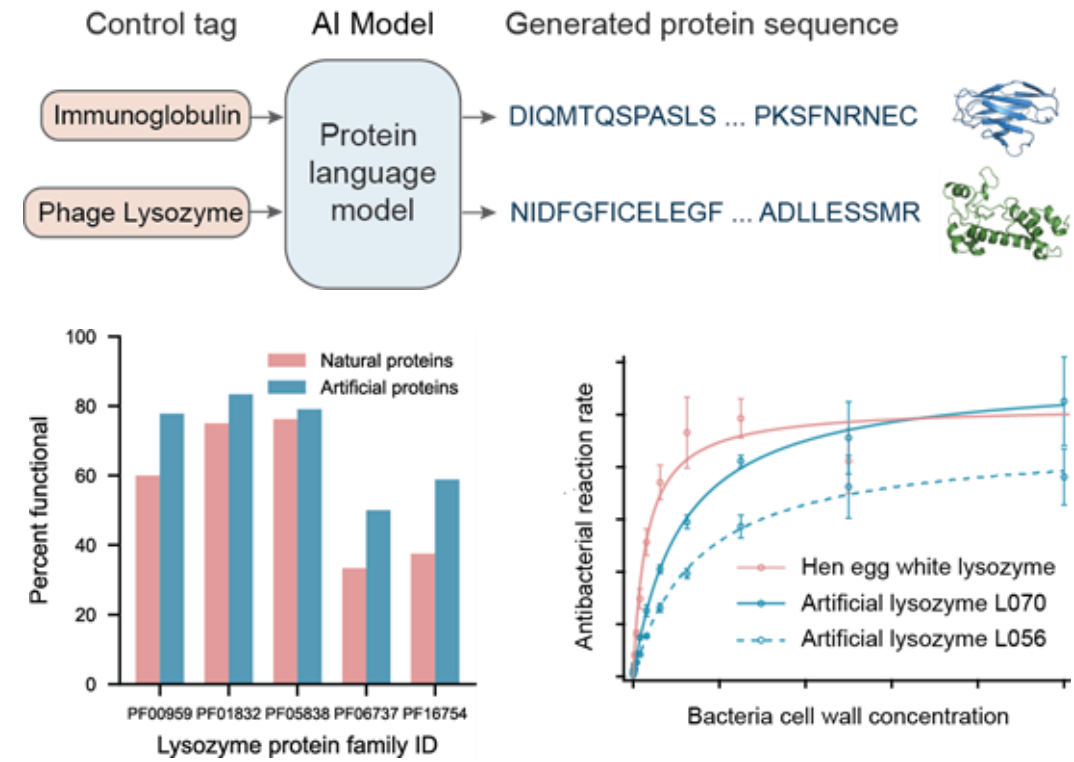
Question Answering (Open-ended)

Cause/Effect

Large language models can generate **functional proteins that are unseen in nature**

► Proteins found in nature today are the product of evolution. But what if AI could generate artificial proteins with useful functionality beyond what evolution has designed?

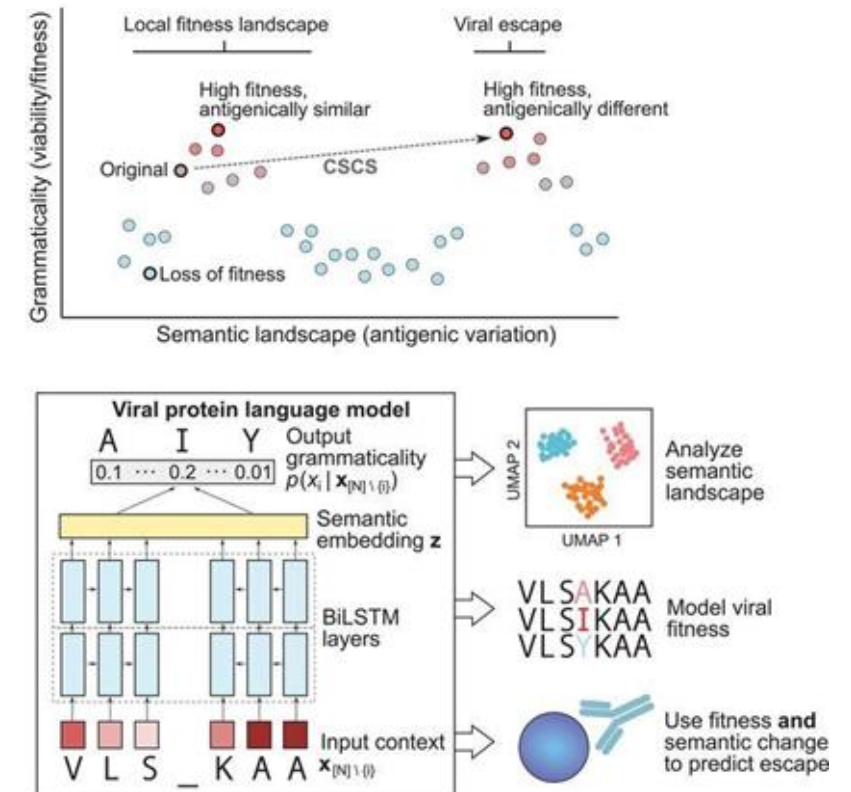
- This work learns a protein language model by predicting the next amino acid for over 280M protein sequences from thousands of protein families (top figure).
- AI-generated proteins across 5 families of antibacterial lysozymes show similar biological performance characteristics as their natural peers, even when their sequence similarity is only 44% (bottom figures).
- The 3D structure of the model-generated artificial lysozyme was then determined by X-ray crystallography showing conserved fold and position of enzyme active site residues compared to the natural protein.



Learning the language of Covid-19 to predict its evolution and escape mutants

► Language models trained on viral sequences can predict mutations that preserve infectivity but induce high antigenic change, akin to preserving “grammaticality” but inducing high “semantic change”.

- Viral escape occurs when a virus mutates to evade neutralizing antibodies from the host immune system. This can impede the development and effectiveness of vaccines, which we’ve seen with the Delta variant.
- Language model evolutionary features help identify the S494P mutation, which decreases the neutralization potential of multiple therapeutic antibodies against SARS-CoV-2 pseudovirus in vitro.
- Going forward, we could imagine vaccine development that corners viral evolution by using language models to better understand how it generates sequence diversity.



Tuning billions of model parameters costs millions of dollars

► Based on variables released by Google et al., you're paying circa \$1 per 1,000 parameters. This means OpenAI's 175B parameter GPT-3 could have cost tens of millions to train. Experts suggest the likely budget was \$10M.

Just how much does it cost to train a model? Two correct answers are “depends” and “a lot”. More quantitatively, here are current ballpark list-price costs of training differently sized BERT [4] models on the Wikipedia and Book corpora (15 GB). For each setting we report two numbers - the cost of one training run, and a typical fully-loaded cost (see discussion of “hidden costs” below) with hyper-parameter tuning and multiple runs per setting (here we look at a somewhat modest upper bound of two configurations and ten runs per configuration).⁴

- \$2.5k - \$50k (110 million parameter model)
- \$10k - \$200k (340 million parameter model)
- \$80k - \$1.6m (1.5 billion parameter model)

For example, based on information released by Google, we estimate that, at list-price, training the 11B-parameter variant⁵ of T5 [5] cost well above \$1.3 million for a single run. Assuming 2-3 runs of the large model and hundreds of the small ones, the (list-)price tag for the entire project may have been \$10 million⁶.

Not many companies – certainly not many startups – can afford this cost. Some argue that this is not a severe issue; let the Googles of the world pre-train and publish the large language models, and let the rest of the world fine-tune them (a much cheaper endeavor) to specific tasks. Others (e.g., Etchemendy and Li [6]) are not as sanguine.

Artificial intelligence / Machine learning

Training a single AI model can emit as much carbon as five cars in their lifetimes

Deep learning has a terrible carbon footprint.

by **Karen Hao**

June 6, 2019

Common carbon footprint benchmarks

in lbs of CO2 equivalent

Roundtrip flight b/w NY and SF (1 passenger)

1,984

Human life (avg. 1 year)

11,023

American life (avg. 1 year)

36,156

US car including fuel (avg. 1 lifetime)

126,000

Transformer (213M parameters) w/
neural architecture search

626,155

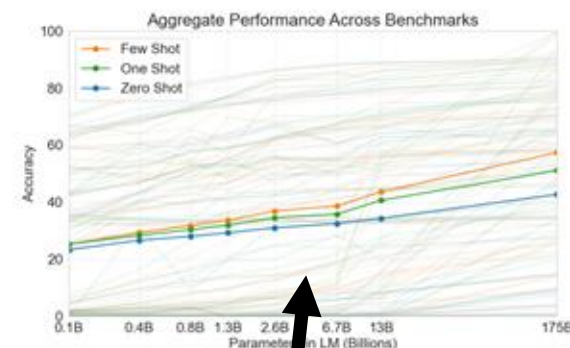


Figure 1.3: Aggregate performance for all 42 accuracy-denominated benchmarks While zero-shot performance improves steadily with model size, few-shot performance increases more rapidly, demonstrating that larger models are more proficient at in-context learning. See Figure 3.8 for a more detailed analysis on SuperGLUE, a standard NLP benchmark suite.

Chart: MIT Technology Review • Source: Strubell et al. • Created with Datawrapper

State of AI Report

October 11, 2022

Definitions

Artificial intelligence (AI): a broad discipline with the goal of creating intelligent machines, as opposed to the natural intelligence that is demonstrated by humans and animals.

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Reinforcement learning (RL): an area of ML in which software agents learn goal-oriented behavior by trial and error in an environment that provides rewards or penalties in response to their actions (called a "policy") towards achieving that goal.

Deep learning (DL): an area of ML that attempts to mimic the activity in layers of neurons in the brain to learn how to recognise complex patterns in data. The "deep" refers to the large number of layers of neurons in contemporary models that help to learn rich representations of data to achieve better performance gains.

Definitions

Model: once a ML algorithm has been trained on data, the output of the process is known as the model. This can then be used to make predictions.

Self-supervised learning (SSL): a form of unsupervised learning, where manually labeled data is not needed. Raw data is instead modified in an automated way to create artificial labels to learn from. An example of SSL is learning to complete text by masking random words in a sentence and trying to predict the missing ones.

(Large) Language model (LM, LLM): a model trained on textual data. The most common use case of a LM is text generation. The term “LLM” is used to designate multi-billion parameter LMs, but this is a moving definition.

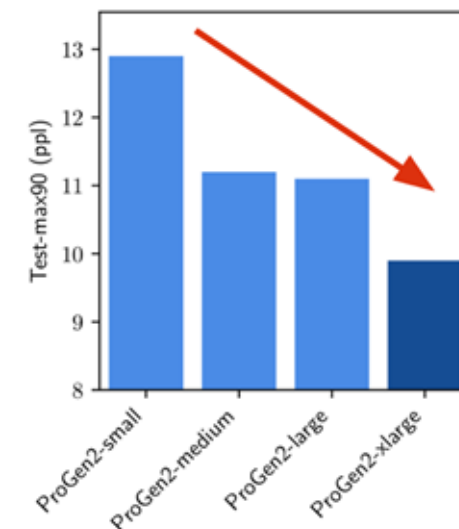
Computer vision (CV): enabling machines to analyse, understand and manipulate images and video.

Transformer: a model architecture at the core of most state of the art (SOTA) ML research. It is composed of multiple “attention” layers which learn which parts of the input data are the most important for a given task. Transformers started in language modeling, then expanded into computer vision, audio, and other modalities.

Language models for proteins: a familiar story of open source and scaled models

► **Researchers independently applied language models to the problems of protein generation and structure prediction while scaling model parameter. They both report large benefits from scaling their models.**

- Salesforce researchers find that scaling their LMs allows them to better capture the training distribution of protein sequences (as measured by perplexity).
- Using the 6B param ProGen2, they generated proteins with similar folds to natural proteins, but with a substantially different sequence identity. But to unlock the full potential of scale, the authors insist that more emphasis be placed on data distribution.
- Meta et al. introduced the ESM family of protein LMs, whose sizes range from 8M to 15B (dubbed ESM-2) parameters. Using ESM-2, they build ESMFold to predict protein structure. They show that ESMFold produces similar predictions to AlphaFold 2 and RoseTTAFold, but is an order of magnitude faster.
- This is because ESMFold doesn't rely on the use of multiple sequence alignments (MSA) and templates like AlphaFold 2 and RoseTTAFold, and instead only uses protein sequences.



Model	# Params	Validation Perplexity	LR P@L	CASP14	CAMEO
ESM-2	8M	10.33	0.17	0.37	0.48
	35M	8.95	0.30	0.41	0.56
	150M	7.75	0.44	0.49	0.65
	650M	6.95	0.52	0.51	0.70
	3B	6.49	0.54	0.52	0.72
	15B	6.37	0.54	0.55	0.72

Mathematical abilities of Language Models largely surpass expectations

► Built on Google's 540B parameter LM PaLM, Google's Minerva achieves a 50.3% score on the MATH benchmark (43.4 pct points better than previous SOTA), beating forecasters expectations for best score in 2022 (13%). Meanwhile, OpenAI trained a network to solve two mathematical olympiad problems (IMO).

- Google trained its (pre-trained) LLM PaLM on an additional 118GB dataset of scientific papers from arXiv and web pages using LaTeX and MathJax. Using chain of thought prompting (including intermediate reasoning steps in prompts rather than the final answer only) and other techniques like majority voting, Minerva improves the SOTA on most datasets by at least double digit pct points.
- Minerva only uses a language model and doesn't explicitly encode formal mathematics. It is more flexible but can only be automatically evaluated on its final answer rather than its whole reasoning, which might justify some score inflation.
- In contrast, OpenAI built a (transformer-based) theorem prover built in the Lean formal environment. Different versions of their model were able to solve a number of problems from AMC12 (26), AIME (6) and IMO (2) (increasing order of difficulty).

Question: Assume that the variance of the first n natural numbers is 10, and the variance of the first m even natural numbers is 16. Compute $m + n$.

Model output: Let σ^2 be the variance of the first m even natural numbers, and τ^2 be the variance of the first n natural numbers. Then, we have $\tau^2 = \frac{n^2-1}{12}$ and $\sigma^2 = \frac{m^2-1}{3}$. We solve the system of equations:

$$\frac{n^2 - 1}{12} = 10$$

$$\frac{m^2 - 1}{3} = 16.$$

This translates into the system

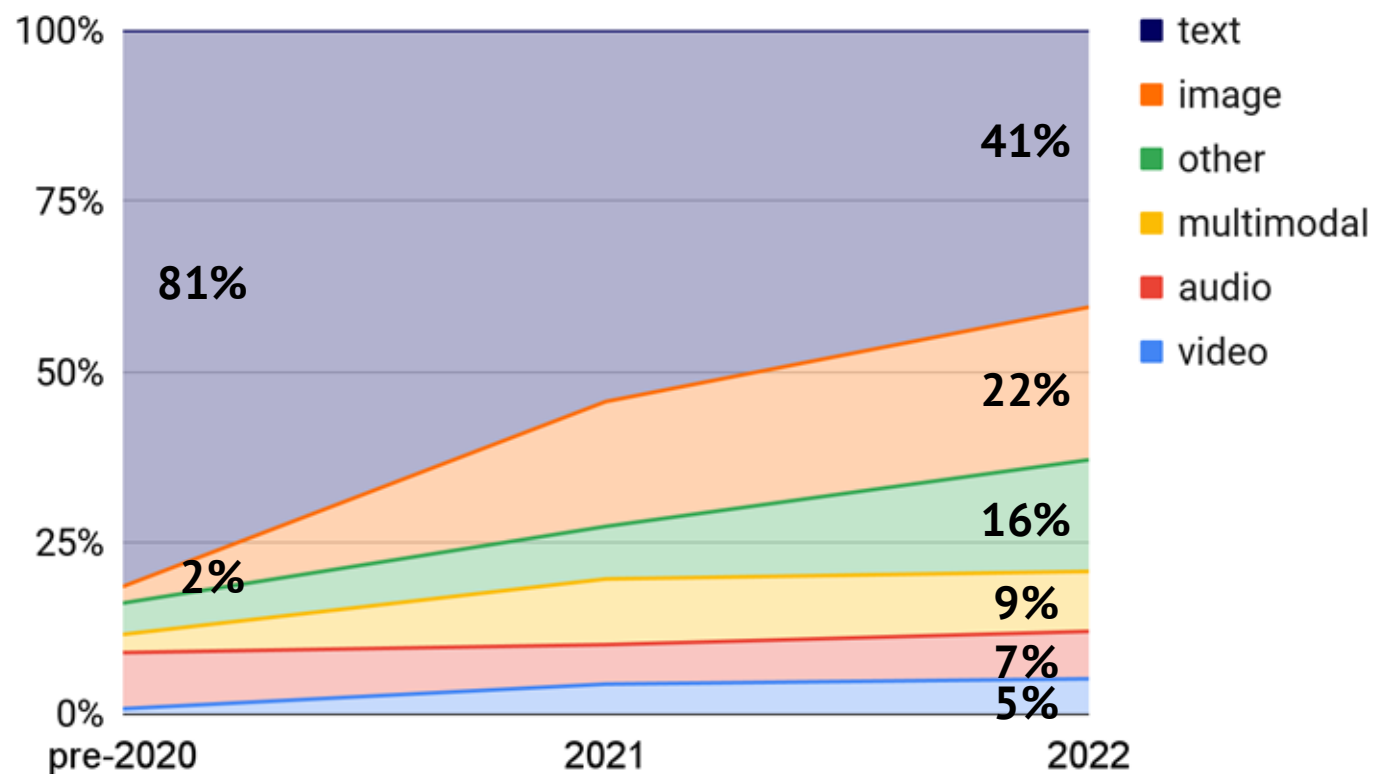
$$n^2 = 121$$

$$m^2 = 48 + 1 = 49.$$

Therefore, $n = \boxed{11}$ and $m = \boxed{7}$, so $n + m = \boxed{18}$.

Transformers are becoming truly cross-modality

- In the 2020 State of AI Report we predicted that transformers would expand beyond NLP to achieve state of the art in computer vision. It is now clear that **transformers are a candidate general purpose architecture**. Analysing transformer-related papers in 2022 shows just how ubiquitous this model architecture has become.



The text-to-image diffusion model frenzy gives birth to new AI labs

- ▶ **Stability.ai and Midjourney came out of seemingly nowhere with text-to-image models that rival those of established AI labs. Both have APIs in beta, Midjourney is reportedly profitable, and Stability has already open-sourced their model. But more on their emergence and research dynamics in our Politics section.**

MIDJOURNEY



DALL-E 2



STABLEDIFFUSION



Behind the scenes of shooting the moon landing, Hollywood studio, 1969, backstage photograph, astronaut actors, lighting

MIDJOURNEY



DALL-E 2



STABLEDIFFUSION



film still, portrait of an old man, wrinkles, dignified look, grey silver hair, peculiar nose, wise, eternal wisdom and beauty, incredible lighting and camera work, depth of field, bokeh, screenshot from a hollywood movie



The text-to-video generation race has started

- Research on diffusion-based text-to-video generation was kicked-off around April 2022, with work from Google and the University of British Columbia. But in late September, new research from Meta and Google came with a jump in quality, announcing a sooner-than-expected DALL-E moment for text-to-video generation.

Make-a-Video



Imagen Video



Phenaki



- Meta made the first splash from Big Tech in text-to-video generation by releasing Make-a-Video, a diffusion model for video generation.

Text, Audio, Image, and Video

LLMs empower robots to execute diverse and ambiguous instructions

► Thanks to their large range of capabilities, LLMs could in principle enable **robots to perform any task by explaining its steps in natural language**. But LLMs have little contextual knowledge of the robot's environment and its abilities, making their explanations generally infeasible for the robot. **PaLM-SayCan solves this**.

- Given an ambiguous instruction “I spilled my drink, can you help?”, a carefully prompt-engineered LLM (e.g. Google’s PaLM) can devise a sequence of abstract steps to pick up and bring you a sponge. But any given skill (e.g. pick up, put down) needs to be doable by the robot in concordance with its environment (e.g. robot sees a sponge).
- To incentivise the LLM to output feasible instructions, SayCan maximises the likelihood of an instruction being successfully executed by the robot.
- Assume the robot can execute a set of skills. Then, for any given instruction and state, the system selects the skill that maximizes: the probability of a given completion (restricted to the set of available skills) times the probability of success given the completion and the current state. The system is trained using reinforcement learning.
- Researchers tested SayCan on 101 instructions from 7 types of language instructions. It was successful in planning and execution 84% and 74% of the time respectively.



State of AI Report

October 12, 2023

Definitions

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AI Agent: an AI-powered system that can take actions in an environment. For example, an LLM that has access to a suite of tools and has to decide which one to use in order to accomplish a task that it has been prompted to do.

AI Safety: a field that studies and attempts to mitigate the risks (minor to catastrophic) which future AI could pose to humanity.

Computer vision (CV): the ability of a program to analyse and understand images and video.

Deep learning (DL): an approach to AI inspired by how neurons in the brain recognise complex patterns in data. The “deep” refers to the many layers of neurons in today’s models that help to learn rich representations of data to achieve better performance gains.

Diffusion: An algorithm that iteratively denoises an artificially corrupted signal in order to generate new, high-quality outputs. In recent years it has been at the forefront of image generation.

Generative AI: A family of AI systems that are capable of generating new content (e.g. text, images, audio, or 3D assets) based on 'prompts'.

Graphics Processing Unit (GPU): a semiconductor processing unit that enables a large number calculations to be computed in parallel. Historically this was required for rendering computer graphics. Since 2012 GPUs have adapted for training DL models, which also require a large number of parallel calculations.

Definitions

(Large) Language model (LM, LLM): a model trained on vast amounts of (often) textual data to predict the next word in a self-supervised manner. The term “LLM” is used to designate multi-billion parameter LMs, but this is a moving definition.

Machine learning (ML): a subset of AI that often uses statistical techniques to give machines the ability to “learn” from data without being explicitly given the instructions for how to do so. This process is known as “training” a “model” using a learning “algorithm” that progressively improves model performance on a specific task.

Model: a ML algorithm trained on data and used to make predictions.

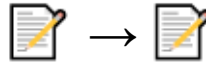
Natural language processing (NLP): the ability of a program to understand human language as it is spoken and written.

Prompt: a user input often written in natural language that is used to instruct an LLM to generate something or take action.

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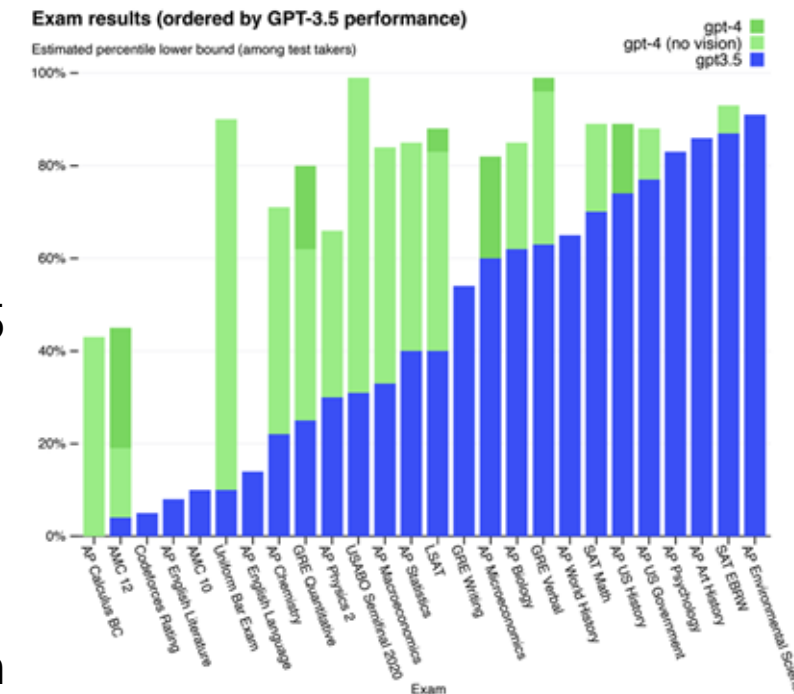
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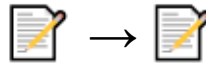


GPT-4 is out and it crushes every other LLM, and many humans

► GPT-4 is OpenAI's latest Large Language Model. In contrast with text-only GPT-3 and follow-ups, GPT-4 is multimodal: it was trained on both text and images; it can among other capabilities generate text based on images. At 8,192 tokens when it was released, it had already exceeded the previous-best GPT-3.5 in possible input size. It is, of course, trained using RLHF. Equipped with these advances, GPT-4 is, as of the release of this report, the uncontested most generally capable AI model.

- OpenAI did a comprehensive evaluation of GPT-4 not only on classical NLP benchmarks, but also on exams designed to evaluate humans (e.g. Bar exam, GRE, Leetcode).
- GPT-4 is the best model across the board. It solves some tasks that GPT-3.5 was unable to, like the Uniform Bar Exam where GPT-4 scores 90% compared to 10% for GPT-3.5. On most tasks, the added vision component had only a minor impact, but it helped tremendously on others.
- OpenAI reports that although GPT-4 still suffers from hallucinations, it is factually correct 40% more often than the previous-best ChatGPT model on an adversarial truthfulness dataset (generated to fool AI models).

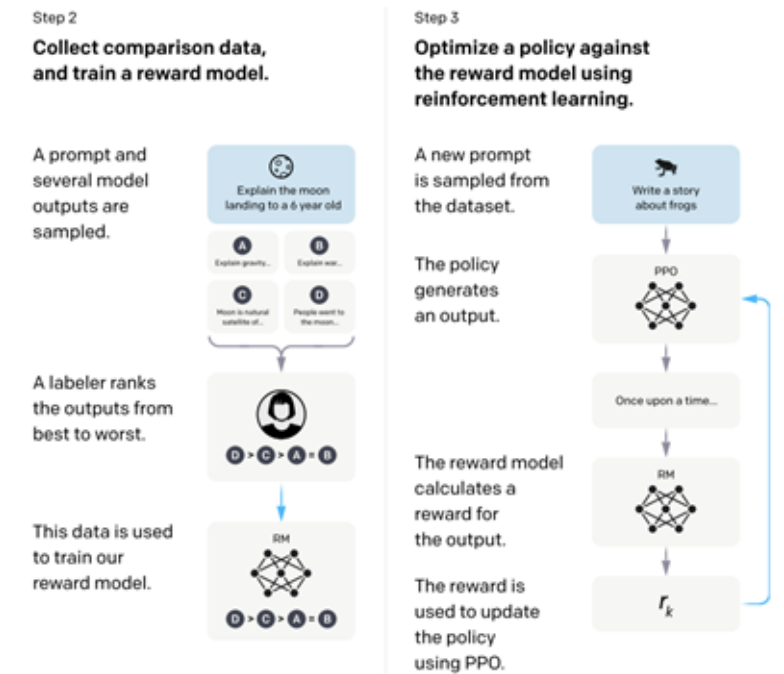




Fueled by ChatGPT's success, RLHF becomes MVP

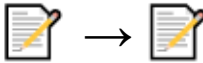
► In last year's Safety section (Slide 100), we highlighted how Reinforcement Learning from Human Feedback (RLHF) – used in InstructGPT – helped make OpenAI's models safer and more helpful for users. Despite a few hiccups, ChatGPT's success proved the technique's viability at a massive scale.

- “RLHF involves humans ranking language model outputs sampled for a given input, using these rankings to learn a reward model of human preferences, and then using this as a reward signal to finetune the language model with using RL.” In its modern form, it dates back to 2017, when OpenAI and DeepMind researchers applied it to incorporate human feedback in training agents on Atari games and to other RL applications.
- RLHF is now central to the success of state of the art LLMs, especially those designed for chat applications. These include Anthropic's Claude, Google's Bard, Meta's LLaMa-2-chat, and, of course, OpenAI's ChatGPT.
- RLHF requires hiring humans to evaluate and rank model outputs, and then models their preferences. This makes this technique hard, expensive, and biased¹. This motivated researchers to look for alternatives.



Typical steps of RLHF, which follow an initial step of supervised fine-tuning of a pre-trained language model, e.g. GPT-3.

¹ We will cover other issues of RLHF in the Safety section.

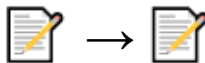


The GPT-4 technical report puts the nail in the coffin of SOTA LLM research...

► OpenAI published a technical report on GPT-4 where it didn't disclose any useful information for AI researchers, signalling the definitive industrialization of AI research. Google's PaLM-2 technical report suffered the same fate, while (OpenAI spinoff) Anthropic didn't bother releasing a technical report for its Claude models.

- “Given both the competitive landscape and the safety implications of large-scale models like GPT-4, this report contains no further details about the architecture (including model size), hardware, training compute, dataset construction, training method, or similar”, OpenAI writes in the GPT-4 technical report published on arXiv.
- When Google released PaLM 2, its most capable LLM, the company wrote in the technical report: “Further details of model size and architecture are withheld from external publication.”
- As the economic stakes and the safety concerns are getting higher (you can choose what to believe), traditionally open companies have embraced a culture of opacity about their most cutting edge research.





...unless LLaMas reverse the trend

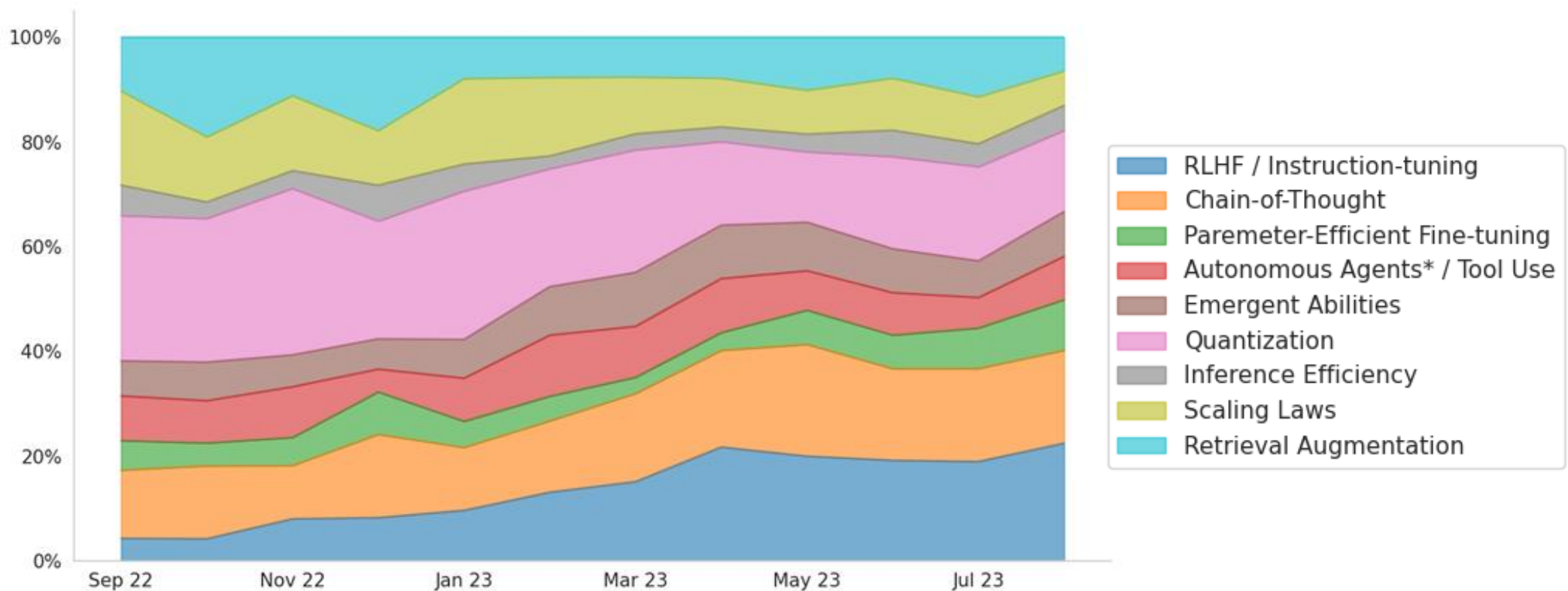
- In February '23, Meta released a series of models called LLaMa. At their release, they stood out as being the most capable models trained exclusively on publicly available datasets. Meta initially granted access to the LLaMa model weights on demand only to researchers, but the weights were quickly leaked and published online.
- The LLaMa-1 models use regular transformers, with slight changes to the architecture. The authors also made a few changes to the optimizer and to the implementation of attention. As a result, *“when training a 65B-parameter model, [their] code processes around 380 tokens/sec/GPU on 2048 A100 GPU with 80GB of RAM. This means that training over [their] dataset containing 1.4T tokens takes approximately 21 days.”*
 - The LLaMa-1 models outperform GPT-3 (the original one, not the InstructGPT variants) and are competitive with DeepMind’s Chinchilla and Google’s PaLM.
 - LLaMa-1 didn’t allow commercial use, prompting heavy criticism around the term “open-source” that Meta used to describe the model release. But a second LLaMa iteration appeased most of the open source community.

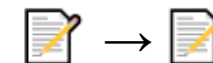
		RACE-middle	RACE-high
GPT-3	175B	58.4	45.5
PaLM	8B	57.9	42.3
	62B	64.3	47.5
	540B	68.1	49.1
LLaMA	7B	61.1	46.9
	13B	61.6	47.2
	33B	64.1	48.3
	65B	67.9	51.6

Table 6: **Reading Comprehension.** Zero-shot accuracy.

Trending topics

- RLHF / Instruction-tuning emerges as the most trending topic since the end of 2022.



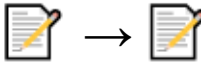


Context length is the new parameter count

► The AI community has extensively verified that when models are trained correctly, their parameter count is a proxy for their capabilities. But these capabilities are sometimes constrained by the size of input that language models can process. Context length has consequently been an increasingly important theme of research.

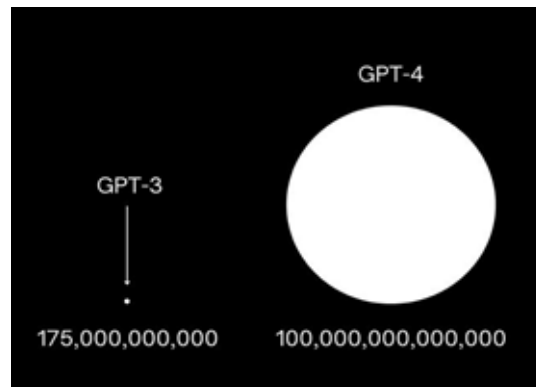
- One of the most alluring promises of LLMs is their few-shot capabilities, i.e. the ability of an LLM to answer a request on a given input without further training on the user's specific use case. But that's hindered by a limited context length due to the resulting compute and memory bottleneck.
- Several innovations have been used to increase the context length of LLMs. Some fundamentally make the memory footprint of attention smaller (FlashAttention). Others enable models to train on small contexts but run inference on larger ones (ALiBi) – this is called length extrapolation – at the price of minimal finetuning and removing positional encodings. Other techniques worth looking into include RoPE and Positional Interpolation.
- Among long-context LLMs: Anthropic's Claude with 100K, OpenAI's GPT-4 with 32K, MosaicML MPT-7B with 65K+, LMSys's LongChat with 16K. But is context all you need?



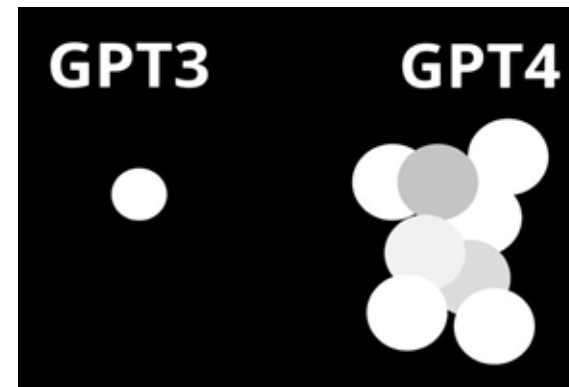


2022 Prediction: language models trained on huge amounts of data

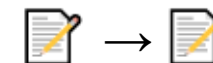
- In 2022, we predicted: “A SOTA LM is trained on 10x more data points than Chinchilla, proving data-set scaling vs. parameter scaling”. Although OpenAI didn’t confirm – and we probably won’t know anytime soon – a sort of consensus seems to be reached among experts about leaked information on the model size, architecture, and the dollar cost of GPT-4. GPT-4 was reportedly trained on ~13 trillion tokens, 9.3x more tokens than Chinchilla.
- The tiny corp founder George Hotz presented the most plausible rumour: “*Sam Altman won’t tell you that GPT-4 has 220B parameters and is a 16-way mixture model with 8 sets of weights*”, and Soumith Chintala, PyTorch co-founder, confirmed. Neither the total size of the model nor using a Mixture of Experts model is unheard of. If the rumours are to be believed, no fundamental innovation underpins GPT-4’s success.



meme



truth?



Are we running out of human-generated data?

- Assuming current data consumption and production rates will hold, research from Epoch AI predicts that “we will have exhausted the stock of low-quality language data by 2030 to 2050, high-quality language data before 2026, and vision data by 2030 to 2060.” Notable innovations that might challenge the hypotheses in the article are speech recognition systems like OpenAI’s **Whisper** that could make all audio data available for LLMs, as well as new OCR models like Meta’s Nougat. It is rumored that plenty of transcribed audio data has already been made available to GPT-4.

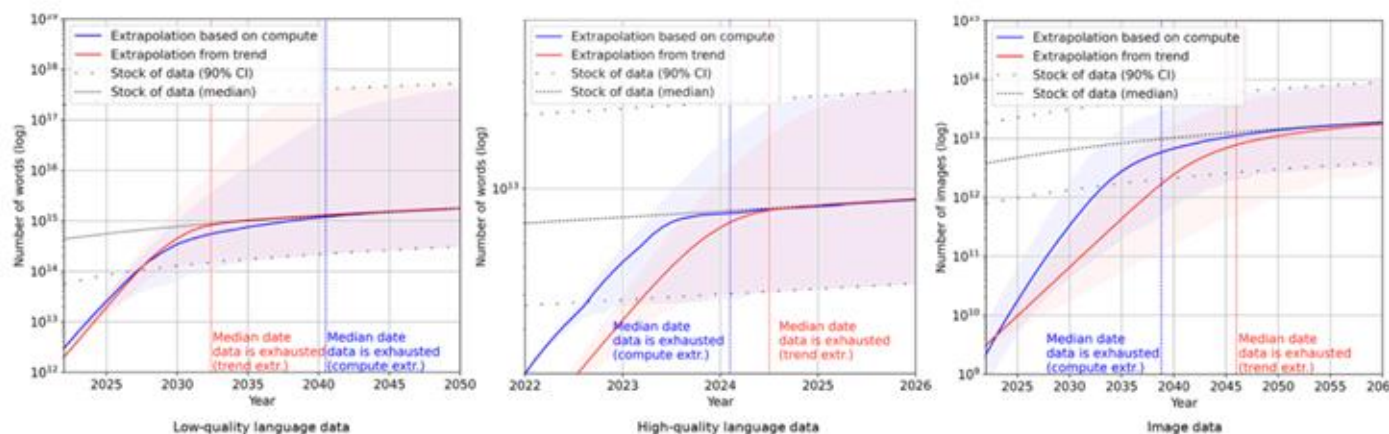
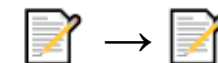


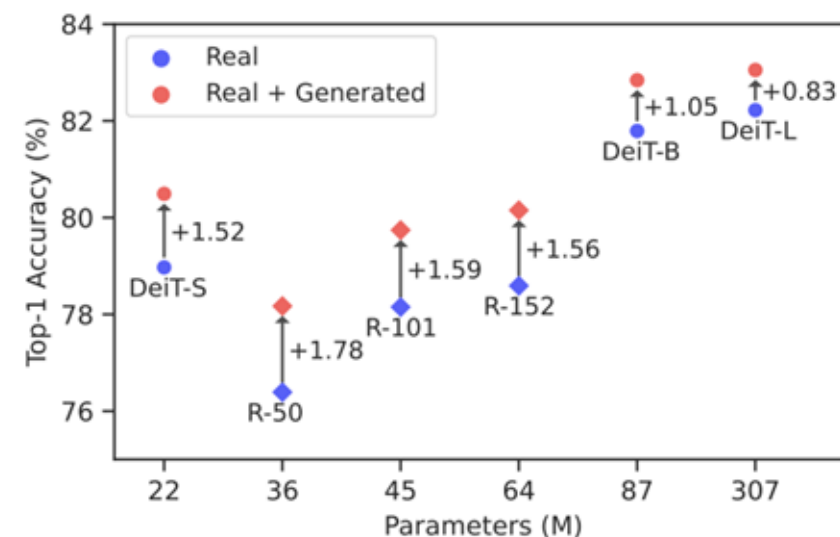
Figure 1: ML data consumption and data production trends for low quality text, high quality text and images.

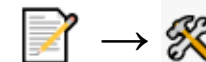


Breaking the data ceiling: AI-generated content

▶ Another perspective that improving generating models open is expanding the pool of available training data via AI-generated content. We're nowhere near a definitive answer: Synthetic data is becoming more helpful, but there is still evidence showing that in some cases generated data makes models forget.

- Despite the seemingly infinitely proprietary and publicly available data, the largest models are actually running out of data to train on, and testing the limits of scaling laws. One way to alleviate this problem (which has been extensively explored in the past) is to train on AI-generated data, whose volume is only bounded by compute.
- Researchers from Google fine-tune the Imagen text-to-image model for class-conditional ImageNet, then generated one to 12 synthetic versions of ImageNet on which they trained their models (in addition to the original ImageNet). They showed that increasing the size of the synthetic dataset monotonically improved the model's accuracy.
- Other researchers showed that the compounding errors from training on synthetic text online may result in model collapse, “*where generated data end up polluting the training set of the next generation of models*”. The way forward might be carefully-controlled data-augmentation (so as usual).

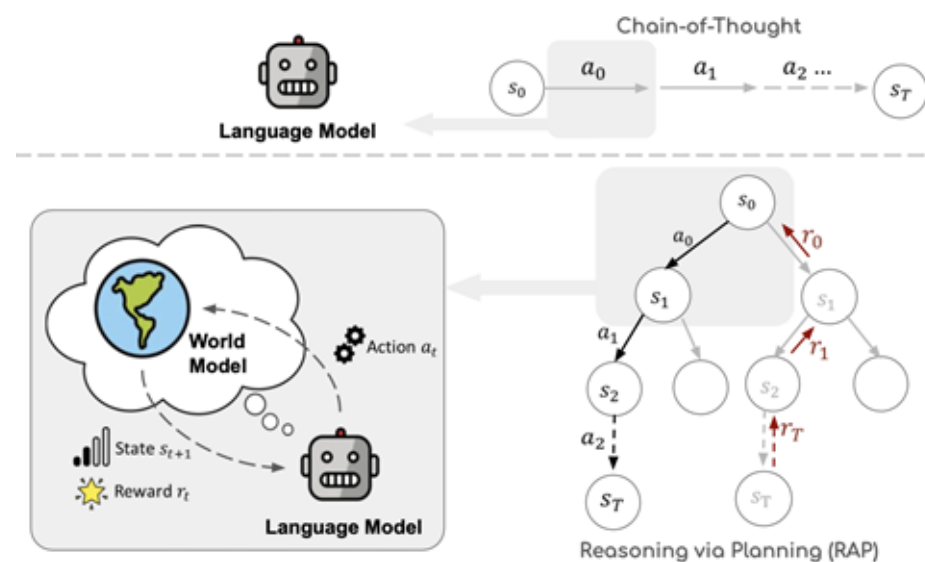




Reasoning with language model is planning with a world model

► Reasoning has been traditionally thought of as searching a space of possible outcomes and picking the best one. By containing so much information about the world, LLMs offer the opportunity of generating this space (often called a world model) in which planning algorithms can explore. Reasoning via Planning (RAP) uses Monte Carlo Tree Search to find a high-reward reasoning path efficiently.

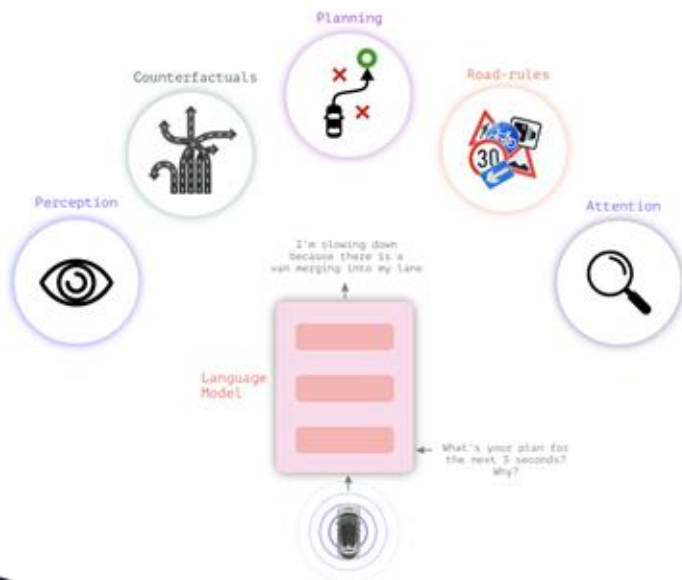
- The world model can generate an action as well as predict the next state reached by taking that action. This produces a reasoning trace which makes the LM more coherent than Chain of Thought methods which predict next actions but not next world states.
- The rewards are also obtained from the LM and used to maintain a state-action value function for planning with MCTS.
- While being significantly more expensive, RAP outperforms Chain-of-Thought reasoning approaches on plan generation, math reasoning and logical reasoning. RAP on LLaMA-33B even outperforms CoT on GPT-4 in a setting of Blocksworld.





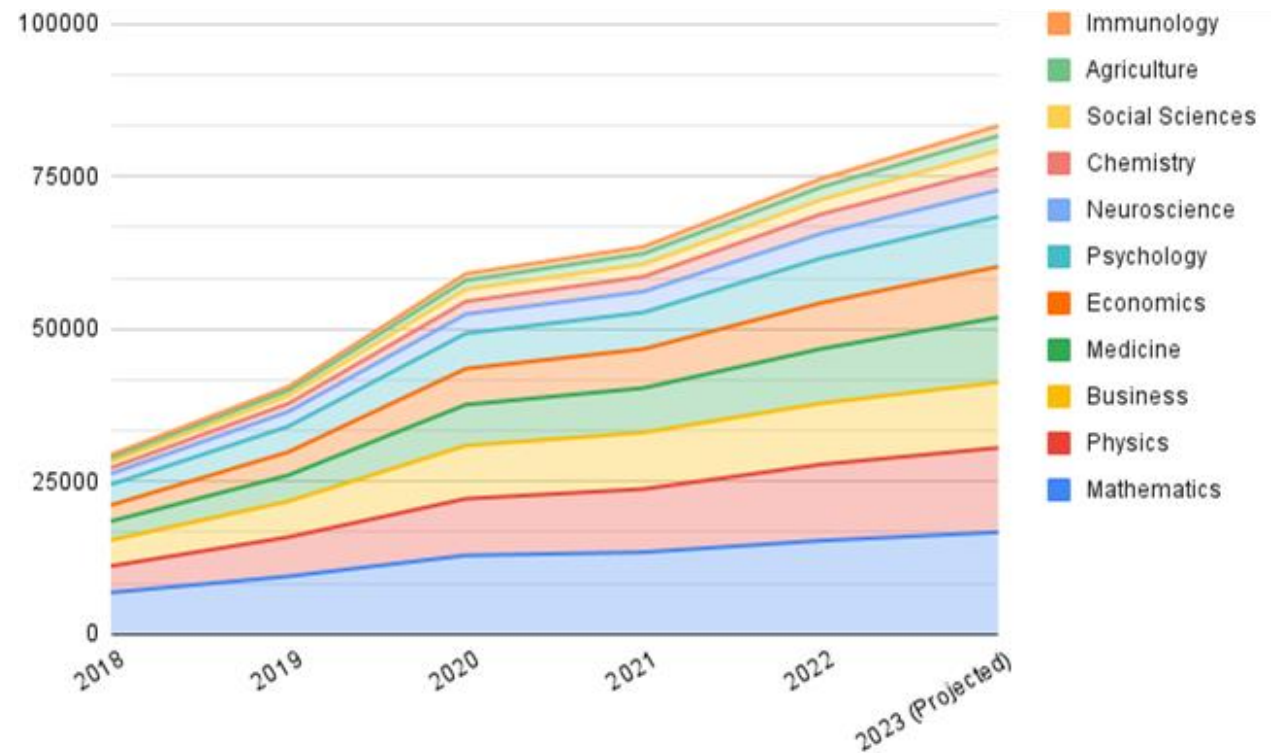
Leveraging LLMs for autonomous driving

- ▶ LINGO-1 is Wayve's vision-language-action model that provides driving commentary, such as information about the driving behaviour or the driving scene. It can also answer questions in a conversational manner. LINGO-1 can be a game changer in terms of explainability of end-to-end driving models as well improve reasoning and planning.



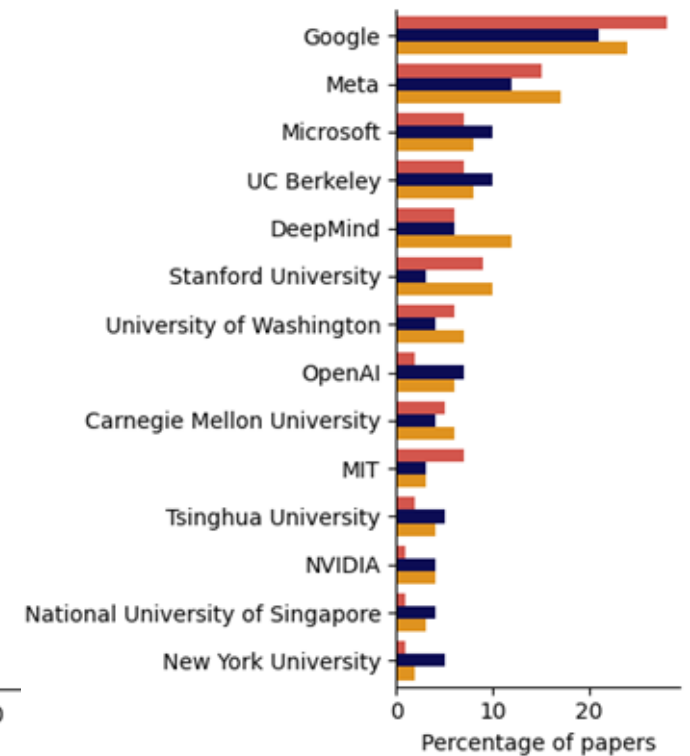
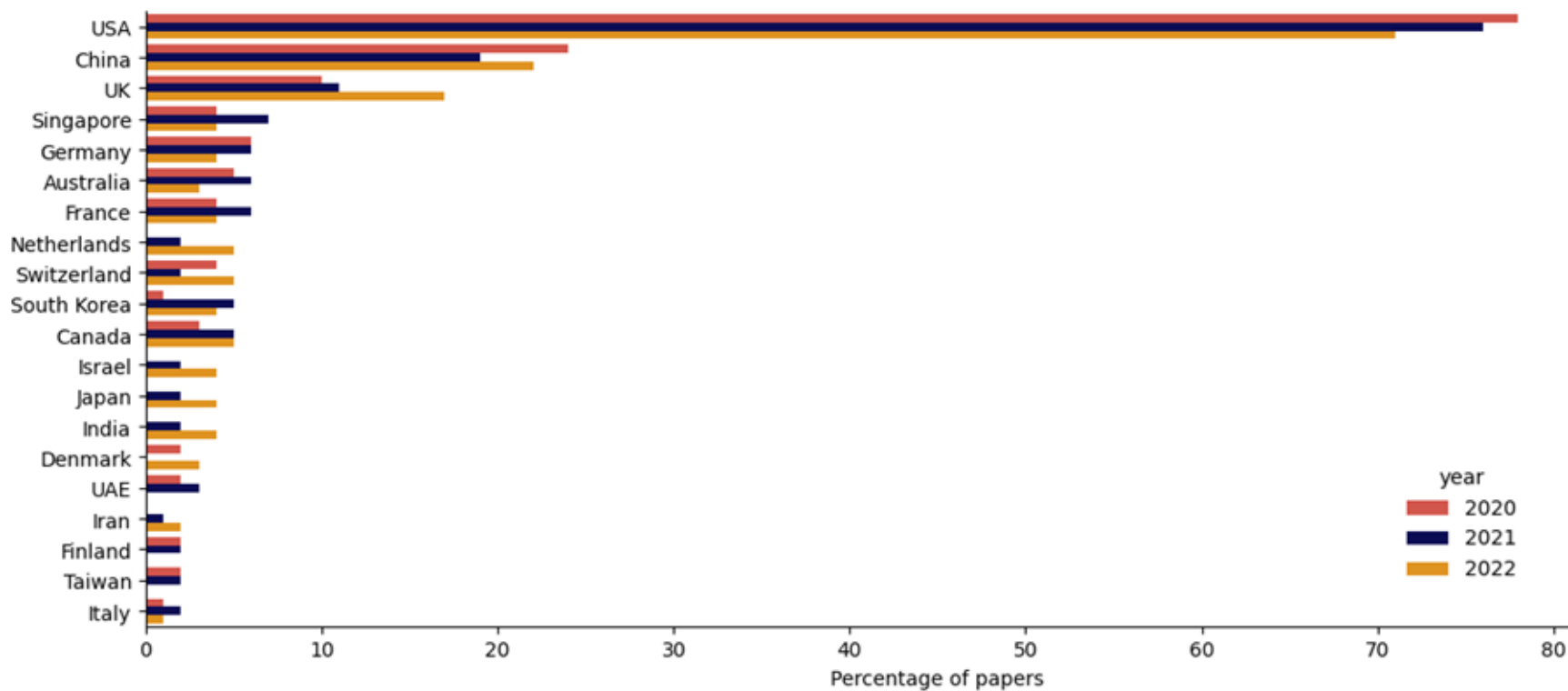
AI for science: medicine is growing fastest but mathematics captures the most attention

- The top 20 scientific fields applying AI to accelerate progress include physical, social, life and health sciences. Out of all the highest increase in the number of publications is Medicine. We expect there to be significant research breakthroughs in the foreseeable future as a result of AI's use in the sciences.



Most impactful research comes from very few places

► >70% of the most cited AI papers in the last 3 years have authors from US-based institutions and organisations.



State of AI Report

October 12, 2024

State of AI Report

October 12, 2025

Outline: W01C

I) An Odyssey

II) Artificial Intelligence (AI)

III) State of AI

III) Data Mining



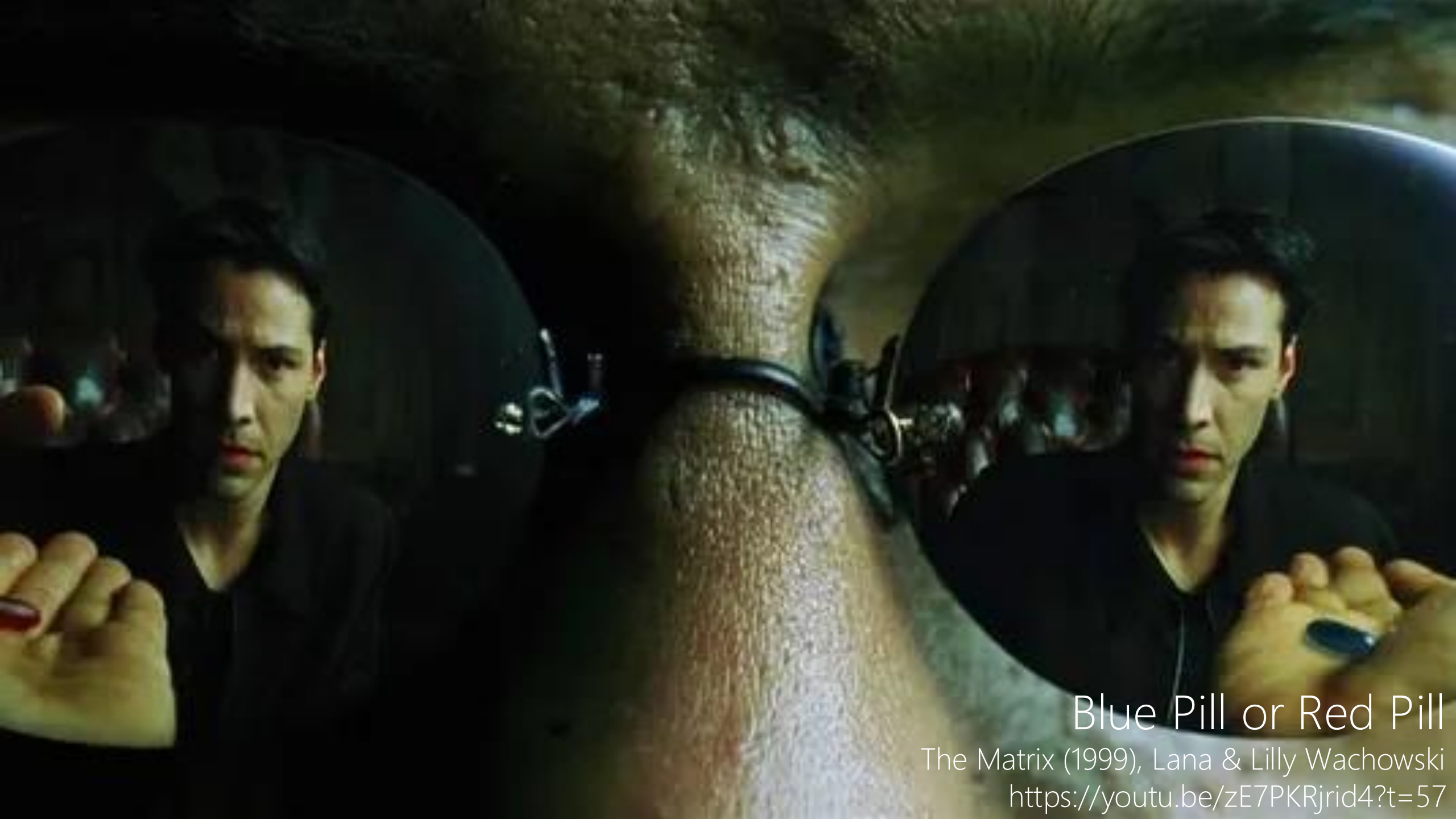
What is Data Mining?

The Matrix (1999), Lana & Lilly Wachowski

<https://www.youtube.com/watch?v=zE7PKRjrid4>



F HD



Blue Pill or Red Pill

The Matrix (1999), Lana & Lilly Wachowski

<https://youtu.be/zE7PKRjrid4?t=57>

Grand Défi (Big Challenge) , Bronze, 175x145x130cm, Nicolas LAVARENNE
<https://www.artseiller.com/en/nicolas-lavarenne-en>



Follow the white rabbit.

Follow the White Rabbit
- The Matrix (1999), Lana & Lilly Wachowski

78

Artificial Intelligence: Let's develop one ...

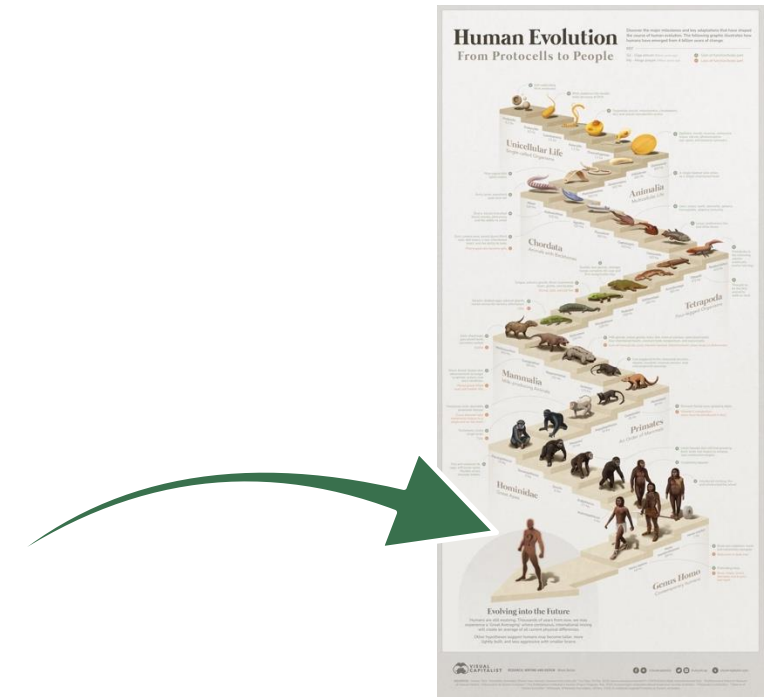
How to emulate human behaviour/actions/... ?

How human obtain behaviour/actions/...?

If I follow the same way that human obtain behaviour/actions/...

Am I going to expose same or similar behaviour/actions/ ...

Am I going to be intelligent?



Artificial Intelligence: Simplify ...

How to emulate **early** human behaviour/actions/... **with less complexity?**

How early human obtain behaviour/actions/...?

If I follow the same way that early human obtain behaviour/actions/...

Am I going to expose same or similar behaviour/actions/ ...

Am I going to be intelligent?



Artificial Intelligence: Simplify more ...

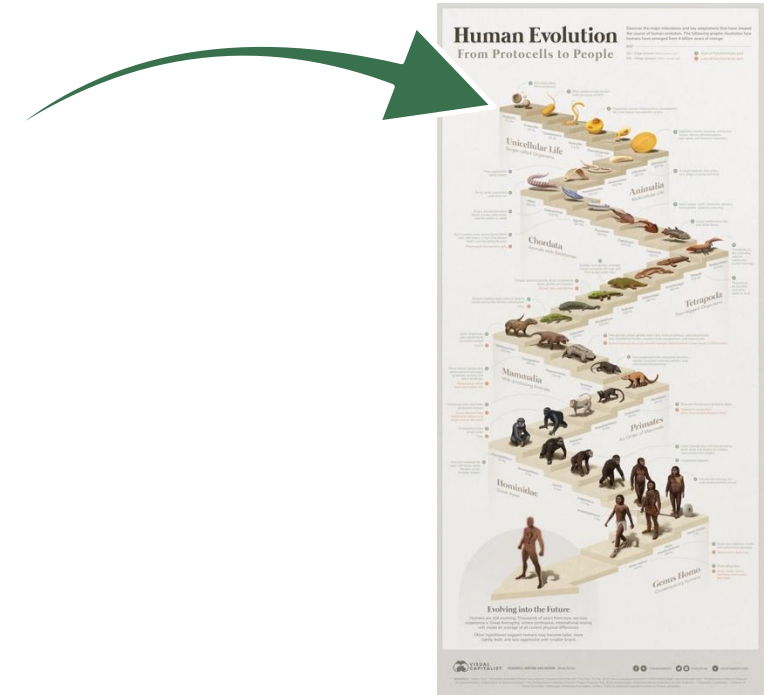
How to emulate **bacteria** behaviour/actions/... **with no complexity?**

How bacteria obtain behaviour/actions/...?

If I follow the same way that bacteria obtain behaviour/actions/...

Am I going to expose same or similar behaviour/actions/ ...

Am I going to be intelligent?



Artificial Intelligence

Example: Machines emulate **bacteria** learning/training to obtain **bacteria-level intelligence**

Escherichia coli (E. coli)

(/ˌɛʃəˈrɪkiə ˈkoʊlaɪ/ ESH-ə-RIK-ee-ə KOH-lye)

They are bacteria, found in many places, environment, foods, water, and the intestines of people and animals.

Most E. coli are harmless and are part of a healthy intestinal tract.

E. coli help us digest food, produce vitamins, and protect us from harmful germs.

1 μm

Artificial Intelligence

Machines emulate bacteria learning/training to obtain bacteria-level intelligence

Bacteria don't have brains, but they can:

Run: Move in a direction

Tumble: Randomly change direction

Are they intelligent, or not intelligent (crazy)?

1 μm



Artificial Intelligence

Machines emulate bacteria learning/training to obtain bacteria-level intelligence

Bacteria don't have brains, but they can:

Run: Move in a direction **for food (nutrients)**

Tumble: Randomly change direction **when fewer nutrients or hitting a barrier**

Are they intelligent, or not intelligent (crazy)?

1 μm

A grayscale microscopic image showing numerous rod-shaped bacteria. In the bottom right corner, there is a white horizontal scale bar with the text "1 μm" above it.

Artificial Intelligence

Machines emulate bacteria learning/training to obtain bacteria-level intelligence

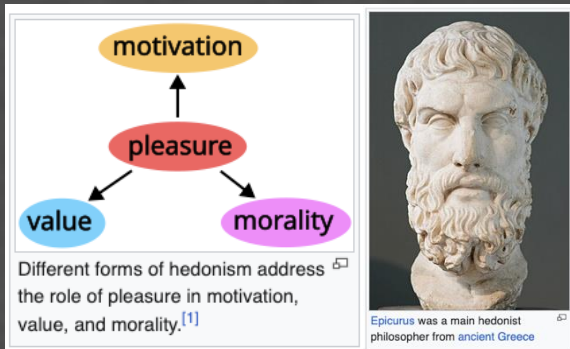
Bacteria don't have brains, but they can:

Run: Move in a direction **for food (nutrients)**

Tumble: Randomly change direction **when fewer nutrients or hitting a barrier**

Are they intelligent, or not intelligent (crazy)?

They want to survive, live longer, grow community, enjoy ...



Hedonism is a family of philosophical views that prioritize pleasure.

1 μm

Artificial Intelligence

Machines emulate bacteria learning/training to obtain bacteria-level intelligence

Am I at food? Boolean (yes/no) check.

```
while(true){  
    if(Run() != food) Tumble();  
}
```

1 μm

A grayscale micrograph showing numerous rod-shaped bacteria. In the bottom right corner, there is a white horizontal scale bar with the text "1 μm" above it.

Artificial Intelligence

Machines emulate bacteria learning/training to obtain bacteria-level intelligence

<https://www.youtube.com/watch?v=-GD0kXgYv2A>

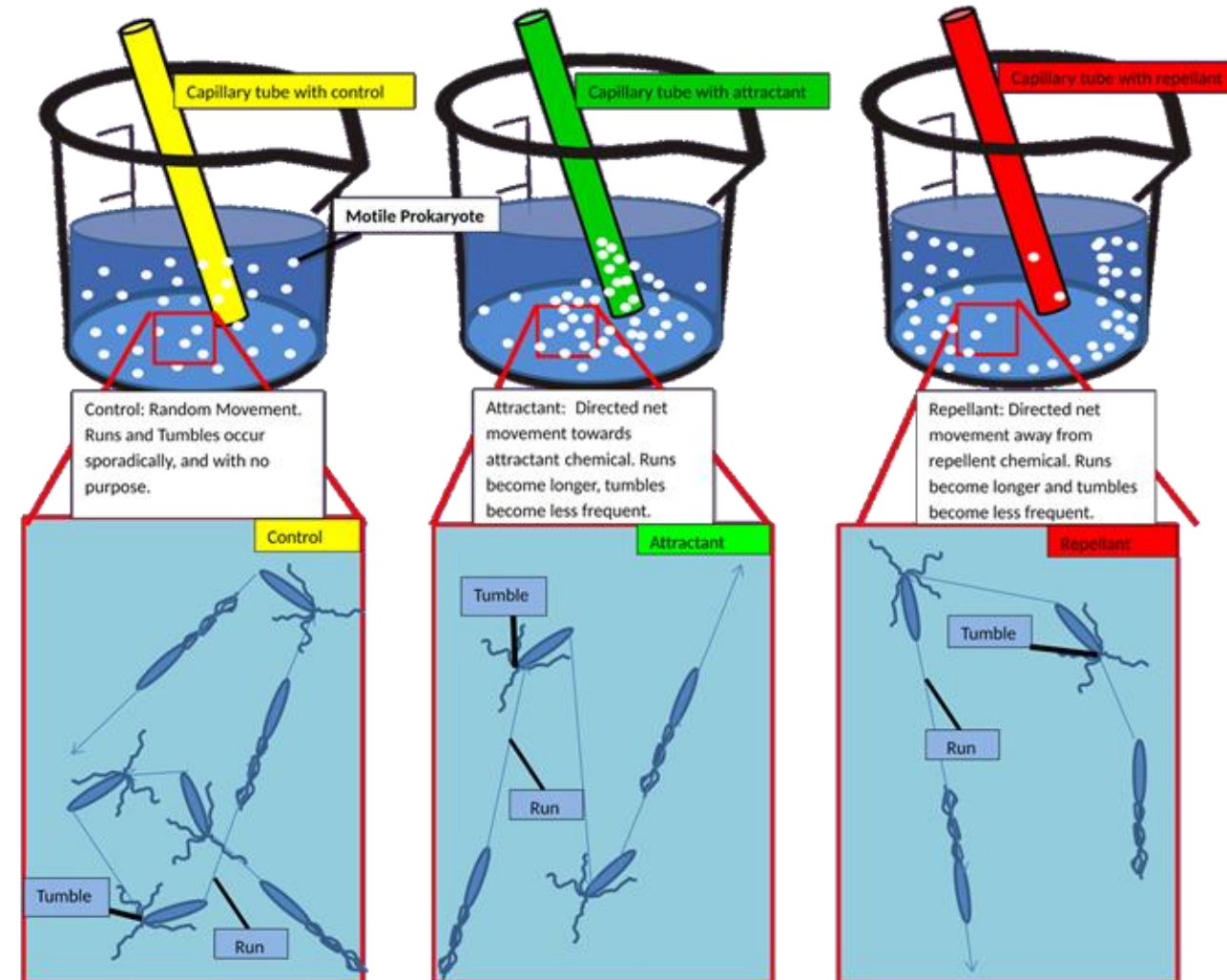
Chemotaxis (from chemo- + taxis)

Movements of an organism
in response to stimulus

Act → move, run, ...

Sense → things get better or worse

Adjust → change direction, tumble, ...



Artificial Intelligence

Machines emulate bacteria learning/training to obtain bacteria-level intelligence

What if we add brains/memory/storage/variable

To record history ...

To record experience ...

1 μm

A grayscale microscopic image showing numerous rod-shaped bacteria. In the bottom right corner, there is a white horizontal scale bar with the text "1 μm" above it.

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Artificial Intelligence: Algorithm (Machine) Learning

Machines emulate bacteria learning/training to obtain bacteria-level intelligence

2026: <https://www.newscientist.com/article/2519284-single-celled-organism-with-no-brain-is-capable-of-pavlovian-learning/>

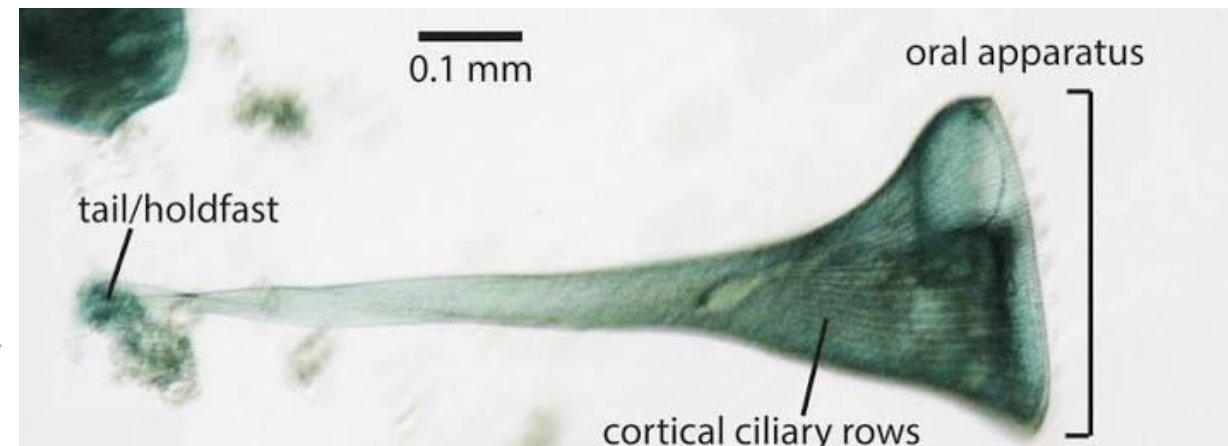
What if we add a single element of brain/memory/bit

To record a single history ...

To record a single experience ...

"been there, done that" → Benefit from experience!

Then, the researchers did two more set-ups: A weak-strong tap sequence (separated by one second) and a weak-weak tap sequence. They found that protists in the first group displayed an enhanced reaction to the first weak tap, as if they perceived that it predicted an accompanying strong tap, until their responses eventually decreased over time as predictive habituation took over. The weak-weak group showed no such enhanced reaction, suggesting that *S. coeruleus* really could learn to associate a weak tap with a forthcoming strong tap.



Stentor Coeruleus STEN-ter see-ROO-lee-us / 'stɛntər sɪ'ruːliəs/

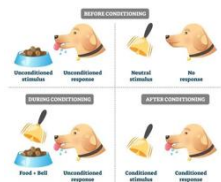
Artificial Intelligence: Algorithm (Machine) Learning

Machines emulate bacteria learning/training to obtain bacteria-level intelligence

2026: <https://www.newscientist.com/article/2519284-single-celled-organism-with-no-brain-is-capable-of-pavlovian-learning/>

Food $\rightarrow$ Lower P $\rightarrow$ Less Tumbling $\rightarrow$ Keep Running.

```
P = 1; //prob of tumbling
while (true) {
    if (Run() == food) P-=0.1;
    if (random() < P) Tumble();//uniform random number in [0, 1]
}
```



Pavlovian Learning: Pavlov's dog Experiment: https://en.wikipedia.org/wiki/Classical_conditioning

The famous experiments by Russian physiologist Ivan Pavlov that led to the discovery of classical conditioning, respondent conditioning, or pavlovian conditioning; A behavioral procedure in which a biologically potent stimulus (e.g. food, a puff of air on the eye) is paired with a neutral stimulus (e.g. the sound of a musical triangle).

Artificial Intelligence: Algorithm (Machine) Learning

Machines emulate **human** learning/training to obtain **human**-level intelligence

What if we add many elements of brain/memory/bit → **Trillions** of bytes (petabytes for human)

To record history ...

To record experience ...

"been there, done that" → Learning from **many** experiences!

→ Learning from data!

→ Data Mining

Artificial Intelligence: Algorithm (Machine) Learning

Machines emulate **human** learning/training to obtain **human**-level intelligence

What if we add many elements of brain/memory/bit → **Trillions** of bytes (petabytes)

To record history ...

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*how does it feel,
how does it feel to be without a home,
like a complete unknown,
like a* 

Artificial Intelligence: Algorithm (Machine) Learning

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*how does it feel,
how does it feel to be without a home,
like a complete unknown,
like a* 

*apple: no English grammar experience
table: no semantic experience
clown: no discourse (poem) experience
rolling stone: no pragmatic (why) experience*

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Artificial Intelligence: Algorithm (Machine) Learning

Machines emulate **human** learning/training to obtain **human**-level intelligence

What if we add a many elements of brain/memory/bit → **Trillions** of bytes (petabytes)

To record history ...

To record experience ...

"been there, done that" → Learning from **many** experiences!

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→ Data Mining

*how does it feel,
how does it feel to be without a home,
like a complete unknown,
like a **rolling stone** — Bob Dylan*

<https://songmeanings.com/songs/view/5730/>



95 Outline: W01 C: Hyperlinks

Constellation Fornax: <https://science.nasa.gov/asset/hubble/hubble-unveils-its-most-colorful-view-of-the-universe-zoom-and-pan/>

A Conversation with HAL: <https://www.youtube.com/watch?v=r13l-TuDcWI>

I'm Sorry, Dave: <https://www.youtube.com/watch?v=Wy4EfdnMZ5g>

Google Duplex: <https://www.youtube.com/watch?v=D5VN56jQMWM>

Beyond the Imitation Game: <https://openreview.net/challenge?redirect=%2Fforum%3Fid%3DuyTL5Bvosj>

Hal Reads Lips: <https://www.youtube.com/watch?v=XDO8OYnmkNY>

LipNet: <https://www.youtube.com/watch?v=fa5QGremQf8&list=PLXkuFIFnXUAPlrXKgtlpctv2NuSo7xw3k&index=2>

AlphaTensor: <https://deepmind.google/blog/discovering-novel-algorithms-with-alphatensor/>

Steve Wozniak: <https://woz.org/>

Ben Goertzel: <https://goertzel.org/>

Nils John Nilsson: <https://robotics.stanford.edu/~nilsson/>

Handle at Boston Dynamics: <https://www.youtube.com/watch?v=7h8mX9ZMs7g>

Physical Intelligence: <https://x.com/svlevine/status/1743328189802598861>

Ex Machina, Knife Scene: <https://www.youtube.com/watch?v=67VATPxULPk>

State of AI Report: <https://www.stateof.ai/>

Pavlov's dog Experiment: https://en.wikipedia.org/wiki/Classical_conditioning

Chemotaxis: <https://www.youtube.com/watch?v=-GD0kXgYv2A>

Single-celled organism with no brain is capable of Pavlovian learning

<https://www.newscientist.com/article/2519284-single-celled-organism-with-no-brain-is-capable-of-pavlovian-learning/>

What's Matrix: <https://www.youtube.com/watch?v=zE7PKRjrid4>

Rolling Stone, Bob Dylan: <https://songmeanings.com/songs/view/5730/>